

Ambient air quality — Standard method for the measurement of the concentration of ozone by ultraviolet photometry

The European Standard EN 14625:2005 has the status of a
British Standard

ICS 13.040.20

National foreword

This British Standard is the official English language version of EN 14625:2005. It supersedes BS ISO 13964:1998 which is withdrawn.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/3, Ambient atmospheres, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
- monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

Cross-references

The British Standards which implement international or European publications referred to in this document may be found in the *BSI Catalogue* under the section entitled “International Standards Correspondence Index”, or by using the “Search” facility of the *BSI Electronic Catalogue* or of British Standards Online.

This publication does not purport to include all the necessary provisions of a contract. Users are responsible for its correct application.

Compliance with a British Standard does not of itself confer immunity from legal obligations.

This British Standard was published under the authority of the Standards Policy and Strategy Committee on 29 March 2005

Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 87 and a back cover.

The BSI copyright notice displayed in this document indicates when the document was last issued.

Amendments issued since publication

Amd. No.	Date	Comments

English version

Ambient air quality - Standard method for the measurement of the concentration of ozone by ultraviolet photometry

Qualité de l'air ambiant - Méthode normalisée de mesurage
de la concentration d'ozone par photométrie UV

Luftqualität - Messverfahren zur Bestimmung von Ozon in
Luft mit dem UV-Verfahren

This European Standard was approved by CEN on 10 December 2004.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the Central Secretariat or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the Central Secretariat has the same status as the official versions.

CEN members are the national standards bodies of Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.



EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

Management Centre: rue de Stassart, 36 B-1050 Brussels

Contents

	Page
Foreword.....	4
1 Scope	5
2 Normative references	5
3 Terms and definitions	5
4 Symbols and abbreviated terms	9
5 Principle.....	12
5.1 General.....	12
5.2 Measuring principle	12
5.3 Type approval test	13
6 Sampling equipment	13
6.1 General.....	13
6.2 Sampling location	13
6.3 Sampling inlet and sampling line.....	14
6.4 Particulate filter.....	14
6.5 Control and regulation of sample flow rate	15
6.6 Sampling pump for the manifold.....	15
7 Analyser equipment	15
7.1 General.....	15
7.2 Interferences	15
7.3 Ultraviolet absorption cell.....	15
7.4 Ultraviolet source lamp	16
7.5 UV detector.....	16
7.6 Ozone-specific scrubber	16
7.7 Switching valve	16
7.8 Temperature indicator	16
7.9 Pressure indicator	16
7.10 Flow rate indicator	17
7.11 Sampling pump for the analyser	17
7.12 Residence time in the sampling system and inside the analyser	17
7.13 Internal ozone span source	17
8 Type approval of ultraviolet photometric ozone analysers.....	17
8.1 General.....	17
8.2 Relevant performance characteristics and performance criteria	17
8.3 Design changes	20
8.4 Procedures for determination of the performance characteristics during the laboratory test	20
8.5 Determination of the performance characteristics during the field test.....	30
8.6 Expanded uncertainty calculation for type approval	34
9 Field operation and ongoing quality control	34
9.1 General.....	34
9.2 Suitability evaluation	34
9.3 Initial installation.....	35
9.4 Ongoing quality control	36
9.5 Calibration of the analyser.....	37
9.6 Checks	38
9.7 Maintenance	41
9.8 Data handling and data reports.....	41
10 Expression of results	42

11	Test reports and documentation.....	42
11.1	Type approval tests	42
11.2	Field operation	42
	Annex A (normative) Correction for ambient nitric oxide	44
	Annex B (normative) Calculation of lack of fit.....	45
	Annex C (informative) Sampling equipment.....	47
	Annex D (informative) Sampling on micro scale.....	49
	Annex E (informative) Ultra violet photometric analyser	50
	Annex F (informative) Manifold testing equipment.....	51
	Annex G (normative) Type approval	52
	Annex H (normative) Calculation of uncertainty in field operation at the hourly alert threshold value.....	74
	Bibliography	86

Foreword

This document (EN 14625:2005) has been prepared by Technical Committee CEN/TC 264 "Air quality", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by September 2005, and conflicting national standards shall be withdrawn at the latest by September 2005.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 Scope

This document specifies a continuous measurement method for the determination of the concentration of ozone present in ambient air based on the ultraviolet photometric measuring principle. This document describes the performance characteristics, and sets the relevant minimum criteria required to select an appropriate ultraviolet photometric ozone analyser by means of type approval tests. It also includes the evaluation of the suitability of an analyser for use in a specific fixed site so as to meet the Directives data quality requirements and requirements during sampling, calibration and quality assurance.

The method is applicable to the determination of the mass concentration of ozone present in ambient air in the range from $0 \mu\text{g}/\text{m}^3$ to $500 \mu\text{g}/\text{m}^3$. This concentration range represents the certification range for the type approval test.

NOTE 1 $0 \mu\text{g}/\text{m}^3$ to $500 \mu\text{g}/\text{m}^3$ of O_3 corresponds to 0 nmol/mol to 250 nmol/mol of O_3 .

The method covers the determination of ambient air concentrations of ozone in zones classified as rural areas, urban and urban-background areas.

NOTE 2 Other ranges may be used for measurement systems applied at rural locations monitoring Ecosystems.

The results are expressed in $\mu\text{g}/\text{m}^3$ (at 20°C and $101,3 \text{ kPa}$).

When the standard is used for other purposes than the EU-directive, the range and uncertainty requirements need not apply.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ENV 13005, *Guide to the expression of uncertainty in measurement*

EN ISO 14956:2002, *Air quality — Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*

ISO 13964:1998, *Air quality — Determination of ozone in ambient air — Ultraviolet photometric method*

3 Terms and definitions

For the purpose of this document, the following terms and definitions apply.

3.1

ambient air

outdoor air in the troposphere excluding workplace air¹

3.2

sample gas temperature

temperature at the sampling inlet outside the monitoring station

¹) As stated in the relevant EU-legislation.

3.3

availability of the analyser

fraction of the total time period for which usually valid measuring data of the ambient air concentration is available from an analyser

3.4

calibration

comparison of the analyser response to a known gas concentration with a known uncertainty

3.5

certification range

concentration range for which the analyser is type approved

3.6

combined standard uncertainty

calculation result of combining the uncertainties determined from all performance characteristics specified in this document according to the prescribed procedures given in this document

3.7

coverage factor

numerical factor used as a multiplier of the combined standard uncertainty in order to obtain an expanded uncertainty

3.8

designated body

body which has been designated for a specific task (type approval tests and/or QA/QC activities in the field) by the competent authority in the Member States

NOTE It is recommended that the designated body is accredited for the specific task according to EN ISO 17025.

3.9

expanded uncertainty

quantity defining an interval about the result of a measurement that may be expected to encompass a large fraction of the distribution of values that could reasonably be attributed to the measurand

NOTE for the purpose of this document the expanded uncertainty is the combined standard uncertainty multiplied by a coverage factor $k = 2$ resulting in an interval with a level of confidence of 95 %

3.10

fall time

difference between the response time (fall) and the lag time (fall)

3.11

independent measurement

individual measurement that is not influenced by a previous individual measurement by separating two individual measurements by at least four response times

3.12

individual measurement

measurement averaged over a time period equal to the response time of the analyser

3.13

influence quantity

quantity that is not the measurand but that affects the result of the measurement (VIM 2.7), either an interferent influence quantity (i.e. the concentration of a substance in the air under investigation that is not the measurand), or an external influence quantity (i.e. a quantity that is not the measurand nor the concentration of a substance in the air mass under investigation)

NOTE Examples are:

- presence of interfering gases in the flue gas matrix (interferent influence quantity);
- temperature of the surrounding air (external influence quantity);

- atmospheric pressure (external influence quantity);
- pressure of the gas sample (external influence quantity).

3.14**interference**

response of the analyser to interferents

3.15**interferent**

component of the air sample, excluding the measured constituent, that effects the output signal

3.16**initial ozone concentration**

ozone concentration in the ambient air just before entrance into the sampling inlet

3.17**lag time**

time interval from the instant at which a step change of sample concentration occurs at the inlet of the analyser to the instant at which the output reading reaches a level corresponding to 10 % of the stable output reading

3.18**lag time (fall)**

lag time for a negative step change

3.19**lag time (rise)**

lag time for a positive step change

3.20**limit value**

level fixed on the basis of scientific knowledge, with the aim of avoiding, preventing or reducing harmful effects on human health and/or the environment as a whole, to be attained within a given period and not to be exceeded once attained

3.21**lack of fit**

maximum deviation of the average of a series of measurements at the same concentration from the linear regression line

3.22**long-term drift**

difference between zero or span readings over a determined period of time (e.g. period of unattended operation)

3.23**monitoring station**

enclosure located in the field in which an ozone analyser has been installed in such a way that its performance and operation comply with the prescribed requirements

3.24**parallel measurement**

measurements from different analysers, sampling from one and the same sampling manifold, starting at the same time and ending at the same time

3.25**performance characteristic**

one of the parameters assigned to equipment in order to define its performance

3.26**performance criterion**

limiting quantitative numerical value assigned to a performance characteristic, to which conformance is tested

3.27

period of unattended operation

time period over which the drift is within the performance criterion for long-term drift

3.28

repeatability (of results of measurement)

closeness of the agreement between the results of successive individual measurements of the same measurand carried out under the same conditions of measurement [1]

NOTE These conditions are called laboratory repeatability conditions and include:

- the same measurement procedure;
- the same observer,
- the same analyser, used under the same conditions;
- at the same location;
- repetition over a short period of time.

3.29

reproducibility under field conditions

closeness of the agreement between the results of simultaneous measurements with two analysers in ambient air carried out under the same conditions of measurement

NOTE 1 These conditions are called field reproducibility conditions and include:

- the same measurement procedure;
- two identical analysers, used under the same conditions;
- at the same monitoring station;
- the period of unattended operation.

NOTE 2 In this document the reproducibility under field conditions is expressed as a value with a level of confidence of 95 %

3.30

residence time inside the analyser

time period for the sampled air to be transported from the inlet of the analyser to the outlet of the absorption cell

3.31

residence time in the sampling system

time period for the sampled air to be transported from the sampling inlet (of the sampling system) to the inlet of the analyser

3.32

response time

time interval from the instant at which a step change of sample concentration occurs at the inlet of the analyser to the instant at which the output reading reaches a level corresponding to 90 % of the stable in output reading

3.33

response time (fall)

response time at a negative step change

NOTE Response time (fall) is the sum of the lag time (fall) and the fall time.

3.34

response time (rise)

response time at a positive step change

NOTE Response time (rise) is the sum of the lag time (rise) and the rise time.

3.35

rise time

difference between the response time (rise) and the lag time (rise)

3.36**sampled air**

ambient air that has been sampled through the sampling inlet and sampling system

3.37**sampling inlet**

entrance to the sampling system where ambient air is collected from the atmosphere

3.38**short-term drift**

difference between zero or span readings at the beginning and end of a 12 h period

3.39**standard uncertainty**

uncertainty of the result of a measurement expressed as a standard deviation

[ENV 13005]

3.40**surrounding temperature**

temperature of the air directly surrounding the analyser (temperature inside the monitoring station or laboratory)

3.41**type approval**

decision taken by a designated body that the pattern of an analyser conforms to the requirements as laid down in this document

3.42**type approval test**

examination of two or more analysers of the same pattern which are submitted by a manufacturer to a designated body; including the tests necessary for approval of the pattern

3.43**uncertainty(of measurement)**

parameter associated with the result of a measurement that characterises the dispersion of the values that could be attributed to the measurement

4 Symbols and abbreviated terms

For the purposes of this document, the following symbols and abbreviated terms apply.

A_a	availability of the analyser
A_v	average concentration of the measurand during the field test
b_{gp}	sensitivity coefficient of the analyser to sample gas pressure change expressed as a percentage of the measured value, obtained during the laboratory type approval test
b_{gt}	sensitivity coefficient of the analyser to sample gas temperature change
b_{st}	sensitivity coefficient of the analyser to surrounding air temperature change
b_v	sensitivity coefficient of the analyser to electrical voltage change
C	O ₃ concentration of the applied gas
C_{P1}	average concentration of the measurements at sampling gas pressure P_1

C_{P2}	average concentration of the measurements at sampling gas pressure P_2
$C_{s,1}$	average concentration of the measurements at span level at the beginning of the drift period
$C_{s,2}$	average concentration of the measurements at span level at the end of the drift period
c_t	test gas concentration
C_{T1}	average concentration of the measurements at sample gas temperature T_1
C_{T2}	average concentration of the measurements at sample gas temperature T_2
C_{V1}	average concentration reading of the measurements at voltage V_1
C_{V2}	average concentration reading of the measurements at voltage V_2
$C_{z,1}$	average concentration of the measurements at zero at the beginning of the drift period
$C_{z,2}$	average concentration of the measurements at zero at the end of the drift period
$C_{\text{const}}^{\text{av}}$	average of at least four independent measurements during the constant concentration period (t_c)
$C_{\text{var}}^{\text{av}}$	average of at least four independent measurements during the variable concentration period (t_v)
$\overline{d_f}$	average difference of parallel measurements
$d_{f,i}$	the i th difference in a parallel measurement
$D_{l,s}$	long-term drift at span concentration c_t
$D_{l,z}$	long-term drift at zero
$D_{s,s}$	short-term drift at span level
$D_{s,z}$	short-term drift at zero
D_{sc}	difference sample/calibration port
E_{ss}	sample system collection efficiency
F_r	response factor in concentration units per voltage output of the analyser
P_1	sampling gas pressure P_1
P_2	sampling gas pressure P_2
R_d	mean analyser response to the test gas directly sampled by the analyser
r_f	reproducibility under field conditions
$r_{f,\text{abs}}$	absolute reproducibility in the field
R_m	mean analyser response to the test gas via the sample manifold
$s_{r,z}$	repeatability standard deviation at zero
$s_{r,ct}$	repeatability standard deviation at concentration c_t

$s_{r,f}$	reproducibility standard deviation under field conditions
s_l	repeatability standard deviation
T	surrounding air temperature
T_1	sample gas temperature T_1
T_2	sample gas temperature T_2
t_d	relative difference between response time (rise) and response time fall
t_f	response time (fall)
t_l	surrounding air temperature at the laboratory
$t_{n-1, 0,05}$	two-sided Students t-factor at a confidence level of 0,05, with $n-1$ degrees of freedom
t_r	response time (rise)
t_t	time period of the field test minus the time for calibration, conditioning and maintenance
t_u	total time period with validated measuring data
t_v	whole number of t_{O_3} and t_{zero} pairs
V_1	minimum voltage V_{min} (V) specified by the manufacturer
V_2	maximum voltage V_{max} (V) specified by the manufacturer
$\bar{X}$	average of measurements
x_1	first average of the measurements at T_1 just after calibration
x_2	second average of the measurements at T_1 just before calibration
x_z	average of the measurements at zero
x_{ct}	average of the measurements at concentration c_t
X_{av}	averaging effect
x_c	average of the measurements using the calibration port
$X_{H_2O,z,ct}$	influence quantity of H_2O with concentration 19 mmol/mol
x_i	the i th measurement
$X_{int,ct}$	influence quantity of the interferent at concentration c_t
$X_{int,z}$	influence quantity of the interferent at zero
X_l	lack of fit (largest residual from the linear regression function)
x_s	average of the measurements using the sample port
X_s	difference between the readings of the recent zero check and the most recent calibration
x_T	average of the measurements at T_{min} or T_{max}

$X_{\text{benx,z,ct}}$	influence quantity of benzene with concentration 1 $\mu\text{mol/mol}$
X_z	difference between the readings of two consecutive zero checks
$(x_{1,i})_i$	the i th measurement result of analyser 1
$(x_{2,i})_i$	the i th measurement result of analyser 2 at the same time as the measurement of analyser 1
Z_1	reading of the first zero check
Z_2	reading of the second zero check
ΔP_m	measured pressure drop induced by the manifold pump
ΔR_a	change in the analyser's response due to the influence of the pressure drop induced by the manifold pump, expressed as a percentage

5 Principle

5.1 General

This document describes the method for measurement of the concentration of ozone in ambient air by means of ultraviolet photometry. The requirements, the specific components, the ultraviolet photometric analyser and its sampling system are described. For the analyser a number of performance characteristics with associated minimum performance criteria are given. The actual values of these performance characteristics for a specific type of analyser have to be determined in a so-called type approval test for which procedures have been described. The type approval test comprises a laboratory and a field test. The selection of a suitable analyser for a specific measuring task in the field is based on the calculation of the combined expanded uncertainty of the measuring method. In this combined expanded uncertainty calculation the actual values of the various performance characteristics of a type approved analyser and the site-specific conditions at the monitoring station are taken into account. The expanded uncertainty of the method shall meet the requirements of the (EU) legislation. For ongoing measurements in the field, requirements and recommendations for quality assurance and quality control are given (see 9.4).

5.2 Measuring principle

Sampled air is drawn continuously through an optical absorption cell where it is irradiated by monochromatic radiation, centred on 253,7 nm, from a stabilised low-pressure mercury (Hg) discharge lamp. The UV radiation, which passes through the absorption cell, is measured by a sensitive photodiode or photomultiplier detector and converted to a measurable electrical signal. Absorption of this radiation by the sampled air within the absorption cell is a measure of the ozone concentration in the ambient air.

Two different systems for the measurement of the ultraviolet absorption by ozone are usually employed.

NOTE In one system the ultraviolet absorption by ozone is determined by means of the difference in ultraviolet absorption between a sample cell and a reference cell (dual-cell type).

In the other system only a single sample cell is employed. The ultraviolet absorption of ozone is determined by alternately supplying sampled air containing ozone to the absorption cell and ozone-free sampled air. Ozone-free sampled air is obtained by passing the sampled air through an ozone catalytic converter in which the ozone is destroyed.

Most modern commercial ozone analysers measure the temperature and pressure of the sampled air in the absorption cell. Using these data an internal microprocessor automatically calculates the measured ozone concentration relative to some chosen reference conditions. For analysers without this automated pressure and temperature compensation, the concentrations need to be corrected manually to the chosen reference conditions.

The concentration of ozone is measured in volume/volume units (if the analyser is calibrated using a volume/volume standard). The final results for reporting are expressed in $\mu\text{g}/\text{m}^3$ using standard conversion factors (see Clause 10).

5.3 Type approval test

The type approval test is based on the evaluation of performance characteristics determined under a prescribed series of tests. In this document test procedures are described for the determination of the actual values of the performance characteristics for at least two analysers in a laboratory and two analysers in the field. These tests shall be performed by a designated body. The type approval of an analyser is based on the calculation of the combined expanded uncertainty in the measurement result derived from the numerical values of the tested performance characteristics and compared to a prescribed maximum uncertainty.

The type approval of an analyser and subsequent OA and QC procedures provide evidence that the defined requirements concerning data quality laid out in relevant EU directives can be satisfied. Appropriate experimental evidence shall be provided by:

- type approval tests performed under conditions of intended use of the specified method of measurement, and
- calculation of expanded uncertainty of results of measurement by reference to ENV 13005.

Field operation and quality control:

Prior to the installation and operation of a type approved analyser at a monitoring station, an expanded uncertainty calculation shall be performed with the actual values of the performance characteristics, obtained during the type approval tests, and the site-specific conditions at that monitoring station. This calculation shall be used to demonstrate the suitability of a type-approved analyser under the actual conditions present at that specific monitoring station.

After the installation of the approved analyser at the monitoring station its correct functioning shall be tested.

Requirements for quality assurance and quality control are given for the operation and maintenance of the sampling system, as well as for the analyser, to ensure that the uncertainty of subsequent measurement results obtained in the field is not compromised.

6 Sampling equipment

6.1 General

Depending on the installation of the ultraviolet photometric analyser at a monitoring station, a single sampling inlet for the analyser may be chosen. Alternatively sampling can take place from a common sampling inlet with a sampling manifold to which other analysers and equipment may be attached. Conditions and layout of the sampling equipment will contribute to the combined expanded uncertainty of the measurement; to minimise this contribution to the combined expanded uncertainty, performance criteria for the sampling equipment are given in the following subclauses.

NOTE In Annex C different arrangements of the sampling equipment are schematically presented.

6.2 Sampling location

The location where the ambient air is sampled and analysed is not specified, as this depends on the measurement requirements (such as measurements in e.g. a rural area or urban-background area).

NOTE For guidance on sampling points on a micro-scale, see Annex D.

6.3 Sampling inlet and sampling line

A specific method for sampling or sampling equipment is not described. It is acceptable that the equipment fulfils the following minimum requirements.

The sampling inlet shall be constructed in such a way that ingress of rainwater into the sampling line (or system) is prevented. The sampling line shall be as short as practicable to minimise the residence time.

The material of the sampling inlet as well as the sampling line (or system) can influence the composition of the sample. In practice, the best materials such as polytetrafluoroethylene (PTFE), perfluoro-ethylene-propylene (FEP), PFA or borosilicate glass shall be used. Copper or copper-based alloys, shall not be used. The influence of the sampling system on the measured concentrations of ozone due to losses shall be less than 2 %.

NOTE The sample line may be moderately heated to avoid any condensation. Condensation may occur in the case of high ambient temperature and/or humidity.

As NO is generally present in the sampled air, a change in the initial ozone concentration will occur due to the reaction of ozone and NO. Therefore the sampling line shall be as short as practicable to minimise the residence time. Calculations have shown that if the residence time is less than 0,5 s then the decay in the initial ozone concentration will be less than 1 % for most ambient ozone and nitric oxide concentrations encountered. In order to avoid a significant change in the concentration of ozone, the total residence time (the sum of the residence time in the sampling system and the residence time inside the analyser) shall be less than 5 s. The effect of residence time on the decay of the initial ozone concentration shall be calculated according to Annex A.

When the calculated loss of initial ozone is more than 5 % and less than 10 %, then corrections shall be made to the measured ozone concentration. When the calculated loss of initial ozone is more than 10 %, measures are to be taken to reduce the residence time (e.g. by shortening of the sampling line or increasing the flow rate through the sampling line).

Depending on the placement of the particulate filter in the sampling system (see 6.4), the sampling line can be contaminated by deposition of dust. This will induce losses of ozone in the sampling line(s). The sampling system shall be cleaned (as stated in 9.4.2) with a frequency which is dependent on the site-specific conditions.

The influence of the pressure drop over the sampling line including the filter shall be such that it causes less than 1 % of the signal output of the analyser.

The sampling line manifold and filter shall be conditioned (at initial installation and after each cleaning) to avoid temporary decrease in the measured ozone concentrations. Both sampling line and filter shall be conditioned with ambient air for a period of at least 30 min at the nominal sample flow rate. The concentrations measured during these periods shall not be included in the calculation of data capture, hourly and annual averages.

These conditioning periods shall not be included in the calculation of the availability of the analyser during the type approval test (8.5.6) and during field operation (11.2.3). Conditioning may also be done before installing in the laboratory.

6.4 Particulate filter

A particulate filter may be placed in this sampling line. The particulate filter shall retain all particles likely to alter the performance of the analyser. It shall be made of PTFE.

NOTE 1 A pore size of the filter of 5 µm usually fulfils this requirement.

NOTE 2 Suitable materials for the filter housing are for example PTFE, PFA, FEP and borosilicate glass.

Aparticulate filter shall be changed periodically depending on the dust loading at the sampling site (as indicated in 8.6.2). The filter housing shall be cleaned at least every six months. Overloading of thisparticulate filter may cause loss of ozone by adsorption on the particulate matter and can increase the pressure drop in the sampling line.

6.5 Control and regulation of sample flow rate

The sample flow rate into the analyser shall be maintained within specifications of the manufacturer of the analyser.

NOTE Flow rate into the UV photometric analyser is usually controlled by means of restrictors.

6.6 Sampling pump for the manifold

When a sampling manifold is used, a pump (or fan) is necessary for sampling ambient air and suction of the sampled air through the sampling manifold. The inlet of the sampling pump (or fan) for the sampling manifold shall be located at the end of the sampling manifold (see Annex C). The sampling pump or fan shall have sufficient rating to ensure that all analysers connected to the manifold are supplied with the required amount of air and to ensure that the residence time of a sample from inlet until entering the analyser is less than 5 s. To verify functioning of this pump, it is recommended to install a flow alarm system. An example of a sampling manifold is given in Annex C.

The influence of the pressure drop induced by the manifold sampling pump on the measured concentration shall be $< 1\%$.

7 Analyser equipment

7.1 General

In Annex E a schematic diagrams is given of one type of ultraviolet photometric ozone analysers.

The ultraviolet photometric analyser consists of the principal components that are described in 7.3 to 7.11.

7.2 Interferences

The UV photometric method is not subject to interference from any of the common gaseous air pollutants in ambient air at concentrations less than 100 nmol/mol. However, there are reported interferences of about 1 nmol/mol in equivalent ozone for a concentration of 125 nmol/mol (v) of sulphur dioxide and 1 nmol/mol in equivalent ozone for a concentration of 500 nmol/mol of nitrogen dioxide.

NOTE Water vapour normally present in the sampled air is the major interferent. From an intercomparison (1994/1995) at ERLAP [17] H₂O interferences (with 85 % RH at 22 °C) were:

- maximum 5,4 % (12 nmol/mol) and minimum 2 % (4,4 nmol/mol) at a level of 220 nmol/mol O₃ and
- maximum 27,4 (9,6 nmol/mol) and minimum 9,7 % (3,4 nmol/mol) at a level of 35 nmol/mol O₃.

NOTE 2 In general it is not expected that Hg concentrations in ambient air (2 ng/m³ to 4 ng/m³) have an interference on the measurement of ozone. However, care and precautions should be taken when elevated Hg concentrations in ambient air are expected. Interferences of Hg have been reported at a level of 10 ng/m³ of Hg resulting in a response of 1 nmol/mol of ozone.

Additionally, there are some reported interferents for analysers using a manganese dioxide ozone scrubber; these are tabulated in Annex A of ISO 13964:1998.

7.3 Ultraviolet absorption cell

The ultraviolet absorption cell shall be constructed of materials inert to ozone, such as fluorocarbon polymer, borosilicate glass, fused silica or fluorocarbon-coated metal. The cell assembly shall be mechanically stable so that vibration or change in the surrounding temperature does not affect the optical alignment. Provisions shall be available for measuring the temperature and pressure of the gas in the ultraviolet absorption cell.

7.4 Ultraviolet source lamp

The UV lamp (e.g. a low-pressure mercury vapour discharge lamp) shall emit monochromatic UV radiation centred at 253,7 nm. The UV lamp shall be electronically stabilized to ensure a constant UV radiance; any variation in UV radiance during the measurement cycles will result in a measurement bias or noise.

When employing a mercury vapour discharge lamp the radiation at 185 nm (which photolyses oxygen to produce ozone) shall be eliminated by means of a high-silica glass enclosure or shield.

NOTE Spectral data show that ambient water vapour should not interfere in the UV wavelength region of interest here (about 200 nm to 300 nm). However, it has been observed in some analysers of one manufacturer that the quartz window of the UV source lamp was covered with a significant number of microscopic scratches. This resulted in a measurable variation in light scattering at the internal window surface due to changing humidity in the sample air. Replacement of such faulty windows eliminated this interference (see [5]).

7.5 UV detector

The optical system shall be designed in such a way that nearly all radiation sensed by the detector is centred at 253,7 nm. The response of this sensor and its associated electronics shall be sufficiently stable so that the analyser meets the required performance criteria.

NOTE Vacuum photodiodes with caesium telluride sensitisation meet this requirement at 253,7 nm and have a negligible sensitivity at other mercury emission wavelengths. Other detectors, such as photomultiplier tubes also satisfy the purpose.

7.6 Ozone-specific scrubber

The ozone-specific scrubber shall contain a material that selectively catalyses the destruction of ozone in the sampled air.

NOTE 1 Manganese dioxide on a metal substrate has been found to do this satisfactorily. Even though this particular compound does not remove any of the other more common pollutant it has been found to partly remove some organic compounds which may be present in the ambient air. Annex A of ISO 13964:1998 lists such interfering chemical compounds.

NOTE 2 A significant decrease in response to ambient ozone may be an indication of scrubber failure. Normally, manufacturers give an average lifetime of such scrubbers; however, the actual lifetime will depend on the sampling location. For example: high pollutant concentrations at an urban-background station may deactivate the scrubber prematurely.

7.7 Switching valve

A switching valve is used to direct the sampled air alternately either through or around the ozone-specific scrubber. It shall be made of a material that is inert to ozone, such as a fluorocarbon polymer.

7.8 Temperature indicator

A temperature-measuring device capable of measuring the sampled air in the ultraviolet absorption cell with a repeatability of ± 2 °C (as expanded uncertainty).

7.9 Pressure indicator

A pressure-measuring device capable of measuring the pressure of the sampled air in the ultraviolet absorption cell with measurement repeatability of 2 kPa (as expanded uncertainty) is required.

The output signal of the analyser is proportional to the density of CO (number of CO molecules) present in the absorption cell and depends on the pressure in the absorption cell. Variations of internal pressure shall be measured and the signal corrected.

7.10 Flow rate indicator

A flow rate indicator shall be included in the analyser. The flow rate shall be measured after passage of the sampled air through the ultraviolet absorption cell.

7.11 Sampling pump for the analyser

The sampling pump is situated at the end of the outlet of the analyser, and draws the sampled air through the analyser. It can be separate or part of the analyser. In any case it shall be capable of operating within the specified flow requirements of the manufacturer of the analyser and pressure conditions required for the UV absorption cell.

7.12 Residence time in the sampling system and inside the analyser

As NO is generally present in the sampled air, a change in the ozone concentration will occur due to the reaction of ozone and NO. In order to avoid a significant change in the concentration of ozone, the total residence time (the sum of the residence time in the sampling system and the residence time inside the analyser) shall be less than 5 s in any case. When the loss of initial ozone is more than 5 % and less than 10 %, then corrections shall be made to the measured ozone concentration, according to Annex A. To this end, the actual nitric oxide concentrations shall be measured and used in the correction formula in Annex A. When the calculated loss of initial ozone is more than 10 %, measures are to be taken to reduce the residence time (e.g. by shortening of the sampling line or increasing the flow rate through the sampling line).

The residence time inside the analyser is the time needed for the sampled air to reach the absorption chamber starting at the inlet of the analyser.

7.13 Internal ozone span source

Some analysers are equipped with an internal ozone span source. This ozone span source shall not be used for calibration purposes. The ozone concentrations generated by this internal span source may be used for functional tests only.

8 Type approval of ultraviolet photometric ozone analysers

8.1 General

The determination of the concentration of ozone in ambient air has to fulfil the requirement of a maximum uncertainty in the measured values, which is prescribed by EU legislation. In order to achieve an uncertainty less than (or equal to) this required uncertainty, the analyser has to fulfil all the criteria for a number of performance characteristics, which are given in this document. The values of the selected performance characteristics shall be evaluated by means of laboratory and field tests. By combining the values of the selected performance characteristics in an uncertainty calculation, a judgement can be made whether or not the analyser meets the criterion of maximum uncertainty prescribed by EU legislation.

This process of assessment (type approval test) of the values of the performance characteristics comprises laboratory and field tests and the calculation of the expanded uncertainty. At least two analysers shall be tested in the laboratory and the same analysers shall be tested during the field test. All tested analysers are required to pass.

A designated body shall perform the type approval tests. The type approval shall be awarded by or on behalf of the competent authority.

NOTE It is recommended that the designated body for the type approval test is accredited for these activities according to EN ISO 17025.

8.2 Relevant performance characteristics and performance criteria

The performance characteristics which shall be determined during a laboratory and field test, and their related performance criteria are given in Table 1.

The determination of the value of the performance characteristics stated in Table 1 shall be performed by a designated body during the laboratory test and the field test according to the procedures described in 8,4 and 8,5.

Table 1 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Lab. test	Field test	Performance criterion for O ₃
1	Repeatability standard deviation at zero	$r_{l,z}$	8.4.5	x		$\leq 1,0$ nmol/mol
2	Repeatability standard deviation at concentration c_t (at a level of the hourly alert threshold)	$r_{l,ct}$	8.4.5	x		$\leq 3,0$ nmol/mol
3	Lack of fit (residual from the linear regression function)		8.4.6			
3a	Largest residual from the linear regression function at concentrations higher than zero	X_l		x		$\leq 4,0$ % of the measured value
3b	Residual at zero	$X_{l,z}$		x		$\leq 5,0$ nmol/mol
4	Sensitivity coefficient of sample gas pressure	b_{sp}	8.4.7	x		$\leq 2,0$ nmol/mol/kPa
5	Sensitivity coefficient of sample gas temperature	b_{gt}	8.4.8	x		$\leq 1,0$ nmol/mol/°C
6	Sensitivity coefficient of surrounding temperature	b_{st}	8.4.9	x		$\leq 1,0$ nmol/mol/°C
7	Sensitivity coefficient of electrical voltage	b_v	8.4.10	x		$\leq 0,30$ nmol/mol/V
8	Interferents at zero and at concentration c_t (at a level of the hourly alert threshold) ^b		8.4.11			
8a	H ₂ O with concentration 19 mmol/mol ^c	$X_{H_2O,z,ct}$		x		≤ 10 nmol/mol
8b	toluene with concentration 0,5 µmol/mol	$X_{tol,z,ct}$		x		$\leq 5,0$ nmol/mol
8c	xylene with concentration 0,5 µmol/mol	$X_{xyl,z,ct}$		x		$\leq 5,0$ nmol/mol
9	Averaging effect	X_{av}	8.4.12	x		$\leq 7,0$ % of the measured value
10	Reproducibility standard deviation under field conditions	r_f	8.5.5		x	$\leq 5,0$ % of the average of a three month period
11	Long-term drift at zero	$D_{l,z}$	8.5.4		x	$\leq 5,0$ nmol/mol
12	Long-term drift at span level ^a	$D_{l,s}$	8.5.4		x	$\leq 5,0$ % of maximum of certification range

No.	Performance characteristic	Symbol	Clause	Lab. test	Field test	Performance criterion for O ₃
13	Short-term drift at zero	$D_{s,z}$	8.4.4	x		$\leq 2,0$ nmol/mol over 12 h
14	Short-term drift at span level ^a	$D_{s,s}$	8.4.4	x		$\leq 6,0$ nmol/mol over 12 h
15	Response time (rise)	t_r	8.4.3	x		≤ 180 s
16	Response time (fall)	t_f	8.4.3	x		≤ 180 s
17	Difference rise time and fall time	t_d	8.4.3	x		≤ 10 % relative difference or 10 s whatever is the greatest
18	Difference sample/calibration port ^d	D_{sc}	8.4.13	x		$\leq 1,0$ %
19	Period of unattended operation		8.5.6		x	3,0 months or less if manufacturer indicates a shorter period, but not less than 2 weeks
20	Availability of the analyser	A_a	8.5.7		x	> 90 %
^a Span level is 70 % to 80% of the certification range ^b Performance criterion is set both at zero and span level ^c an H ₂ O-concentration of 19 mmol/mol is about 80 %RH at 20 °C and 101,3 kPa ^d if relevant						
NOTE nmol/mol = ppb						

8.3 Design changes

The manufacturer shall report any (design) change of the previous type approved monitor to the responsible authority. The underlying idea of the (design) change, as well as the expected consequences with respect to the performance characteristics as stated in Table 1 shall be reported by the manufacturer. The responsible authority shall decide whether or not new tests are necessary. Excluded from reporting obligation are:

- material of the instrument frame (housing);
- electronic connectors;
- bolts, nuts, cable binders and other fixing materials.

8.4 Procedures for determination of the performance characteristics during the laboratory test

8.4.1 General

A designated body shall perform the determination of the performance characteristics in the laboratory as a part of the type approval test. The quality of the materials and equipment used in the described test procedures shall be in accordance with the requirements given in this document. The tests shall be performed on at least two analysers in the laboratory test.

8.4.2 Test conditions

8.4.2.1 General

Before operating the analyser, the operating instructions of the manufacturer shall be followed particularly with regard to the set-up of equipment and the quality and quantity of the consumable products necessary.

The analyser should be allowed to warm up during the time specified by the manufacturer before undertaking any tests. If the warm-up time is not specified, a minimum of 4 h is recommended.

When applying test gases to the analyser, the test gas system shall be operated sufficiently long before starting the tests in order to stabilise the concentrations applied to the analyser.

Most analyser systems are able to give an output signal as a moving average over an adjustable period of time and some systems automatically change this integration time as a function of the frequency of the fluctuations in concentration of the detected pollutant. These options are typically used in order to smooth output data. It needs to be demonstrated that the set value for the averaging time or the use of an active filter will not influence the result of the averaging test and response time test.

During laboratory and field tests for the type approval the settings of the monitor shall be as the manufacturer requires. All settings shall be noted down in the test report.

8.4.2.2 Parameters

During the test for each individual performance characteristic, the values of the following parameters shall be stable within the specified range given in Table 2.

Table 2 — Set points and stability of test parameters

Parameter	Set points	Stability
Sample gas pressure	Manufacturer's specification (Except for the sample gas pressure test, see 8.4.7)	$\pm 0,2$ kPa
Sample gas temperature	20 °C to 23 °C (except for the sample gas temperature test, see 8.4.8)	± 2 °C
Surrounding temperature air	20 °C to 23 °C (except for the temperature of the surrounding air test, see 8.4.9)	± 2 °C
Electrical voltage ^a	At nominal line voltage and within manufacturer's specifications (except for the voltage dependence test, see 8.4.10)	± 1 %
Sample flow to the analyser	Manufacturer's specification	± 1 %
^a For an analyser operating on direct current the type approval test of voltage variation shall be carried out over the range of ± 10 % of the nominal voltage.		

8.4.2.3 Calibration gases and method

For the determination of the various performance characteristics, test gases (air containing a certain O₃ concentration) traceable to national standards shall be used unless otherwise stated in this document. Various methods for the generation of test gases are given in Table 3.

Table 3 — Methods for preparation of test gases and calibration

Method	Description	Traceability/ Standard to be used
Generation of an ozone calibration gas	Preparation by means of an ozone generator	ISO 13964:1998, 5.2 point b
Calibration	Using a primary UV reference photometer	ISO 13964:1998, 6.2/6.2.1/6.2.2 and Clause 7/7.1

The uncertainties in zero and span gases used for the laboratory and field tests shall be proven to be insignificant. Possible contamination of zero and span gas shall not significantly influence the results of the laboratory tests. Therefore the test gases and zero gas shall meet the following specifications:

Maximum permitted uncertainties in the concentration of gases used for laboratory tests: $\pm 3\%$.

Specification of the purity of zero gas and the air feeding the ozon generator (expressed as absolute value) are given in Table 4.

Table 4 — Specification for purity of zero air and the air feeding the ozon generator

Pollutant	Concentration
xylene	< 1 nmol/mol
toluene	< 1 nmol/mol
water	< 150 $\mu\text{mol/mol}$

NOTE It is advised to perform the field tests with the same set of span gas and zero air generators, exclusively reserved for the tests. The stability of both the zero air and the span gas should be guaranteed over a period longer than the test period.

8.4.3 Response time

8.4.3.1 General requirements

The response time of the analyser shall be determined at the nominal sample flow rate specified by the manufacturer.

The sample flow rate shall be kept constant within the requirements as given in 8.4.2 ($\pm 1\%$) during the test.

8.4.3.2 Test procedure

The determination of the response time shall be carried out by applying to the analyser a step function in the concentration from less than 20 % to about 80 % of the maximum of the certification range and vice versa (see Figure 1).

The change from zero gas to span gas needs to be made almost instantly with the use of a suitable valve. The valve outlet shall be mounted direct to the inlet of the analyser, and both zero gas and span gas shall have the same "oversupply" which is vented with the use of a tee. The gas flows of both zero gas and span gas shall be chosen in such a way that the dead time in the valve and tee can be neglected compared to the lag time of the analyser system. The step change is made by switching the valve from zero gas to span gas. This event needs to be timed and is the start ($t = 0$) of the (rise) lag time according to Figure 1. When the reading of 98 % of the applied concentration has been reached, the span gas can be changed to zero again, this event is the start ($t = 0$) of the (fall) lag time. When the reading of 2 % of the applied concentration has been reached the whole cycle as shown in Figure 1 is complete.

The elapsed time (response time) between the start of the step change and reaching 90 % of the analyser final stable reading of the applied concentration shall be measured. The whole cycle shall be repeated four times. The average of the four response times (rise) and the average the four response times (fall) shall be calculated.

The relative difference in response times shall be calculated according to:

$$t_d = \left| \frac{t_r - t_f}{t_r} \right| \times 100\% \quad (1)$$

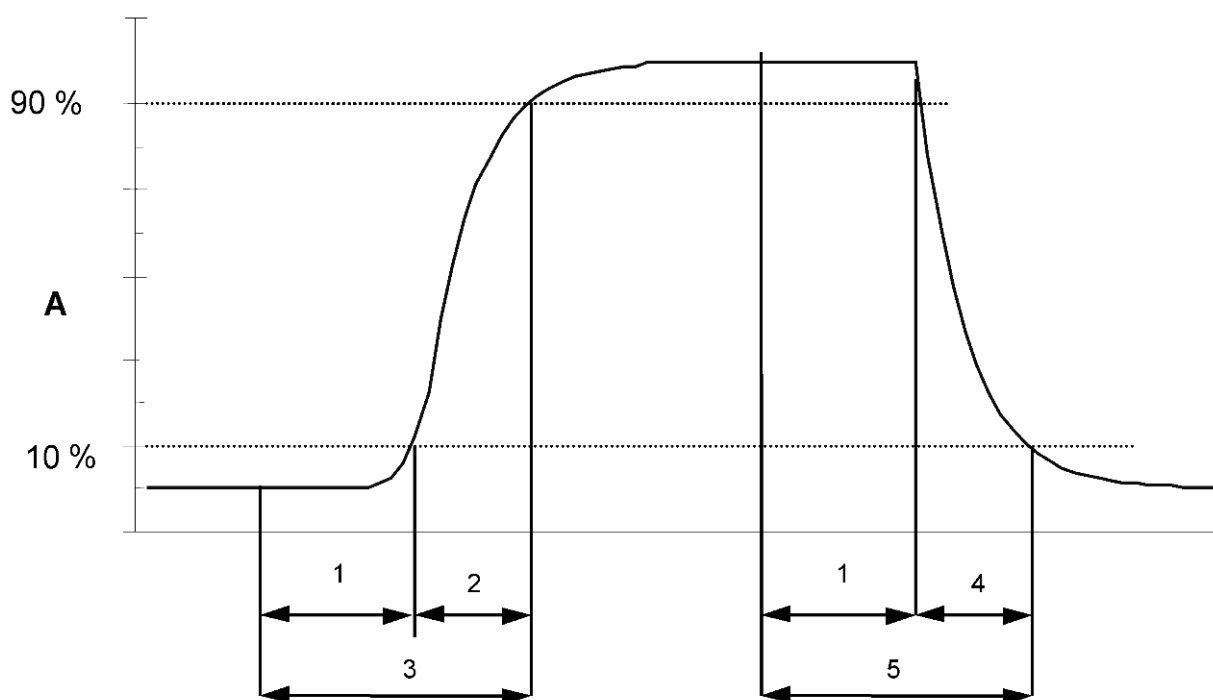
where

t_d is the relative difference between response time (rise) and response time (fall) (%);

t_r is the response time (rise) (average of 4 measurements) (s);

t_f is the response time (fall) (average of 4 measurements) (s).

t_r , t_f and t_d shall comply with the performance criterion in Table 1.



Key

- A Analyser response
- 1 Lag time
- 2 Rise time
- 3 Response time (rise)
- 4 Fall time
- 5 Response time (fall)

Figure 1 — Diagram illustrating the response time

8.4.4 Short-term drift

After the required stabilisation period (8.4.2.1), the analyser shall be adjusted at zero and span level (around 70 % to 80 % of the maximum of the certification range). Wait the time equivalent to one independent reading and then record 20 individual measurements first at zero and then at span concentration. From these 20 measurements the average is calculated for zero and span level.

The analyser shall be kept running under the laboratory conditions (8.4.2.2) whilst analysing ambient air. After a period of 12 h zero and span gas is fed to the analyser. Wait the time equivalent to one independent reading and then record 20 individual measurements first at zero and then at span concentration. The averages for zero and span level shall be calculated.

NOTE Compliance with this test shows that drift is not the dominant factor in any of the test results.

The short-term drift at zero and span level shall be calculated as follows:

$$D_{s,z} = (C_{z,2} - C_{z,1}) \quad (2)$$

where

$D_{s,z}$ is the 12-hour drift at zero (nmol/mol);

$C_{z,1}$ is the average of the zero gas measurements at the beginning of the drift period (just after calibration) (nmol/mol);

$C_{z,2}$ is the average of the zero gas measurements at the end of the drift period (12 h) (nmol/mol).

$D_{s,z}$ shall comply with the performance criterion in Table 1.

$$D_{s,s} = (D_{s,2} - D_{s,1}) - D_{s,z} \quad (3)$$

where

$D_{s,s}$ is the 12-hour drift at span (nmol/mol);

$C_{s,1}$ is the average of the span gas measurements at the beginning of the drift period (just after calibration) (nmol/mol);

$C_{s,2}$ is the average of the span gas measurements at the end of the drift period (12 h) (nmol/mol);

$D_{s,s}$ shall comply with the performance criterion in Table 1.

8.4.5 Repeatability standard deviation

After waiting the time equivalent of one independent reading 20 individual measurements both at zero concentration and at a test concentration (c_t) similar to the hourly alert threshold value shall be performed. From these measurements the repeatability standard deviation (s_r) at zero concentration and at concentration c_t shall be calculated according to:

$$s_r = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}} \quad (4)$$

where

s_r is the repeatability standard deviation (nmol/mol);

x_i is the i th measurement (nmol/mol);

$\bar{x}$ is the average of the 20 measurements (nmol/mol);

n is the number of the measurements, $n = 20$.

The repeatability standard deviation shall be calculated separately for both series of measurements (zero gas and concentration c_t).

s_r shall comply with the performance criterion in Table 1, both at zero ($s_{r,z}$) and at the test concentration c_t (hourly alert threshold) ($s_{r,ct}$).

8.4.6 Linearity (lack of fit)

The lack of fit of the analyser shall be tested over the range 0 % to 95 % of the maximum of the certification range using at least six concentrations (including the zero point). The analyser shall be adjusted at a concentration of about 90 % of the maximum of the certification range. At each concentration (including zero), at least five individual measurements shall be performed.

The concentrations shall be applied in the following sequence: 80 %, 40 %, 0 %, 60 %, 20 % and 95 %. After each change in concentration, at least four response times shall be taken into account before the next measurement is performed.

The uncertainty in the dilution ratios for the applied concentrations shall be less than 1,5 % with respect to each other.

NOTE The design of the test (number of concentrations and number of repetitions) is such that the deviation from a linear function can be determined with sufficient accuracy. The test is also sufficiently robust to detect the case of non-linearity in the range from zero to some concentration at the lower end of the range as well as non-linearity from that lower end of the range to the higher end of the range.

Calculation of the linear regression function and residuals shall be performed according to Annex B. The residuals from the linear regression function shall fulfil the criterion as stated in Table 1.

The largest value of the relative residuals is reported as X_l and shall be taken into account in demonstrating compliance with type approval requirement 1. The value of the relative residual value at the level of the hourly limit value shall be taken in the calculation of type approval requirement 2 and 4.

8.4.7 Sensitivity coefficient to sample gas pressure

Measurements are taken at a concentration of about 70 % to 80 % of the maximum of the certification range at an absolute pressure of $80 \text{ kPa} \pm 0,2 \text{ kPa}$ and at an absolute pressure of $110 \text{ kPa} \pm 0,2 \text{ kPa}$. At each pressure after waiting the time equivalent to one independent reading three individual measurements are recorded. From these 3 measurements the averages at each pressure are calculated.

Measurements at different pressures shall be separated by at least four response times.

The sample gas pressure influence is calculated by:

$$b_{gp} = \left| \frac{(C_{P_1} - C_{P_2})}{(P_2 - P_1)} \right| \quad (5)$$

where

b_{gp} is the sample gas pressure influence (nmol/mol/kPa);

C_{P_1} is the average of the measurements at sample gas pressure P_1 (nmol/mol);

C_{P_2} is the average of the measurements at sample gas pressure P_2 (nmol/mol);

P_1 is the sample gas pressure P_1 (kPa);

P_2 is the sample gas pressure P_2 (kPa).

b_{gp} shall comply with the performance criterion in Table 1.

8.4.8 Sensitivity coefficient to sample gas temperature

For the determination of the dependence of the sample gas temperature measurements shall be performed at sample gas temperatures of $T_1 = 0\text{ °C}$ and $T_2 = 30\text{ °C}$. The temperature dependence shall be determined at a concentration of about 70 % to 80 % of the maximum of the certification range. Wait the time equivalent to one independent reading and then record 3 individual measurements at each temperature.

The sample gas temperature measured at the inlet of the analyser shall be held constant for at least 30 min.

The influence of sample gas temperature is calculated from:

$$b_{gt} = \frac{C_{T2} - C_{T1}}{(T_2 - T_1)} \quad (6)$$

where

b_{gt} is the sample gas temperature influence (nmol/mol/K);

C_{T1} is the average concentration of the measurements at sample gas temperature T_1 (nmol/mol);

C_{T2} is the average concentration of the measurements at sample gas temperature T_2 (nmol/molK);

T_1 is the sample gas temperature T_1 (K);

T_2 is the sample gas temperature T_2 (K).

b_{gt} shall comply with the performance criterion in Table 1.

8.4.9 Sensitivity coefficient to surrounding temperature

The influence of the surrounding temperature shall be determined at the following temperatures (within the specifications of the manufacturer):

- 1) at the minimum temperature $T_{\min} = 273\text{ K}$;
- 2) at the laboratory temperature (T_l), see 8.4.2.2;
- 3) at the maximum temperature $T_{\max} = 303\text{ K}$.

For these tests a climate chamber is necessary.

The influence shall be determined at a concentration around 70 % to 80 % of the maximum of the certification range. At each temperature setting, after waiting the time equivalent to one independent reading, 3 individual measurements at zero and at span shall be recorded.

At each temperature setting the criteria for warm-up or stabilisation time are to be met according to 8.4.2.1.

The measurements shall be performed in the following sequence of the temperature settings:

T_l , $T_{\min}$, T_l and T_l , $T_{\max}$, T_l .

At the first temperature (T_l) the analyser shall be adjusted at zero and at span level (70 % to 80 % of the maximum of the certification range). Then three individual measurements are recorded after waiting the time equivalent to one independent reading at T_l , at $T_{\min}$ and again at T_l . This procedure shall be repeated at the temperature sequence of T_l , $T_{\max}$, and at T_l .

In order to exclude any possible drift due to factors other than temperature the measurements at T_i are averaged, which is taken into account in the following formula for calculation of the surrounding air temperature dependence:

$$b_{st} = \left| \frac{x_T - \frac{x_1 + x_2}{2}}{T - T_i} \right| \quad (7)$$

where

- b_{st} is the surrounding air temperature dependence at zero or span and at T_{min} or T_{max} (nmol/molK);
- x_T is the average of the measurements at T_{min} or T_{max} (nmol/mol);
- x_1 is the first average of the measurements at T_i just after calibration (nmol/mol);
- x_2 is the second average of the measurements at T_i just before calibration (nmol/mol);
- T_i is the surrounding air temperature at the laboratory (K);
- T is the surrounding air temperature T_{min} or T_{max} (K).

For reporting the surrounding air temperature dependence the highest value is taken of the two calculations of the temperature dependence at T_{min} and T_{max} .

b_{st} shall comply with the performance criterion in Table 1.

8.4.10 Sensitivity coefficient to electrical voltage

The sensitivity coefficient of electrical voltage shall be determined at both ends of the specified voltage range, V_{min} and V_{max} at zero concentration and at a concentration around 70 % to 80 % of the maximum of the certification range. After waiting the time equivalent to one independent reading three individual measurements at each voltage and concentration level shall be recorded.

The voltage dependence is calculated from:

$$b_v = \frac{(C_{v2} - C_{v1})}{(V_2 - V_1)} \quad (8)$$

where

- b_v is the voltage influence (nmol/mol/V);
- C_{v1} is the average concentration of the measurements at voltage V_1 (nmol/mol);
- C_{v2} is the average concentration of the measurements at voltage V_2 (nmol/mol);
- V_1 is the minimum voltage V_{min} (V) specified by the manufacturer;
- V_2 is the maximum voltage V_{max} (V) specified by the manufacturer.

For reporting the dependence on voltage the highest value of the result at zero and span level shall be taken.

b_v shall comply with the performance criterion in Table 1.

For an analyser operating on direct current the type approval test of voltage variation shall be carried out over the range of ± 10 % of the nominal voltage.

8.4.11 Interferences

The analyser response to certain interferents which are expected to be present in ambient air shall be tested. The interferents can give a positive or negative response. The test shall be performed at zero and at a test concentration (c_t) similar to the hourly alert threshold value.

The concentration of the mixtures of the test gases with the interferent shall have an uncertainty of less than 5 % and shall be traceable to national standards. The interferents to be tested and their respective concentrations are given in Table 1. The influence of each interferent shall be determined separately. A correction on the concentration of the measurand shall be made for the dilution effect due to addition of an interferent (e.g. water vapour).

After adjustment of the analyser at zero and span level the analyser shall be fed with a mixture of zero gas and the interferent to be investigated with the concentration as given in Table 1. With this mixture one independent measurement followed by two individual measurements shall be carried out. This procedure shall be repeated with a mixture of the measurand at concentration c_t and the interferent to be investigated. The influence quantity at zero and concentration c_t are calculated from:

$$X_{\text{int,z}} = x_1 \quad (9)$$

$$X_{\text{int,ct}} = x_2 - c_t \quad (10)$$

where

$X_{\text{int,z}}$ is the influence quantity of the interferent at zero (nmol/mol);

$X_{\text{int,ct}}$ is the influence quantity of the interferent at concentration c_t (nmol/mol);

x_1 is the average of the measurements at zero (nmol/mol);

x_2 is the average of the measurements at concentration c_t (nmol/mol);

c_t is the concentration of the applied gas at the level of the hourly alert threshold value (nmol/mol).

The influence quantity of the interferences shall comply with the performance criteria in Table 1, both at zero and at concentration c_t .

8.4.12 Averaging test

The averaging test gives a measure of the uncertainty in the averaged values caused by short-term concentration variations in the sampled air shorter than the time scale of the measurement process in the analyser. In general the output of an analyser is a result of the determination of a reference concentration (normally zero) and the actual concentration, which takes a certain time.

- For the determination of the uncertainty due to the averaging, the following concentrations are applied to the analyser and readings are taken at each concentration (see Figure 2):
- a stepwise varied concentration of O_3 between zero and concentration c_t (70 % to 80 % of the maximum of the certification range).

The time period (t_c) of the constant O_3 concentration shall be at least equal to a period necessary to obtain four independent readings (which equals at least 16 response times). The time period (t_v) of the varying O_3 concentration shall be at least equal to a period to obtain four independent readings. The time period (t_{O_3}) for the O_3 concentration shall be 45 s followed by a period (t_{zero}) of 45 s of zero concentration.

Further:

c_t is the test concentration (nmol/mol);

t_v is a whole number of t_{O3} and t_{zero} pairs, and contains a minimum of three such pairs.

The change from t_{O3} to t_{zero} shall be within 0,5 s. The change from t_c to t_v shall be within one response time of the analyser under test.

The averaging effect (X_{av}) is calculated according to:

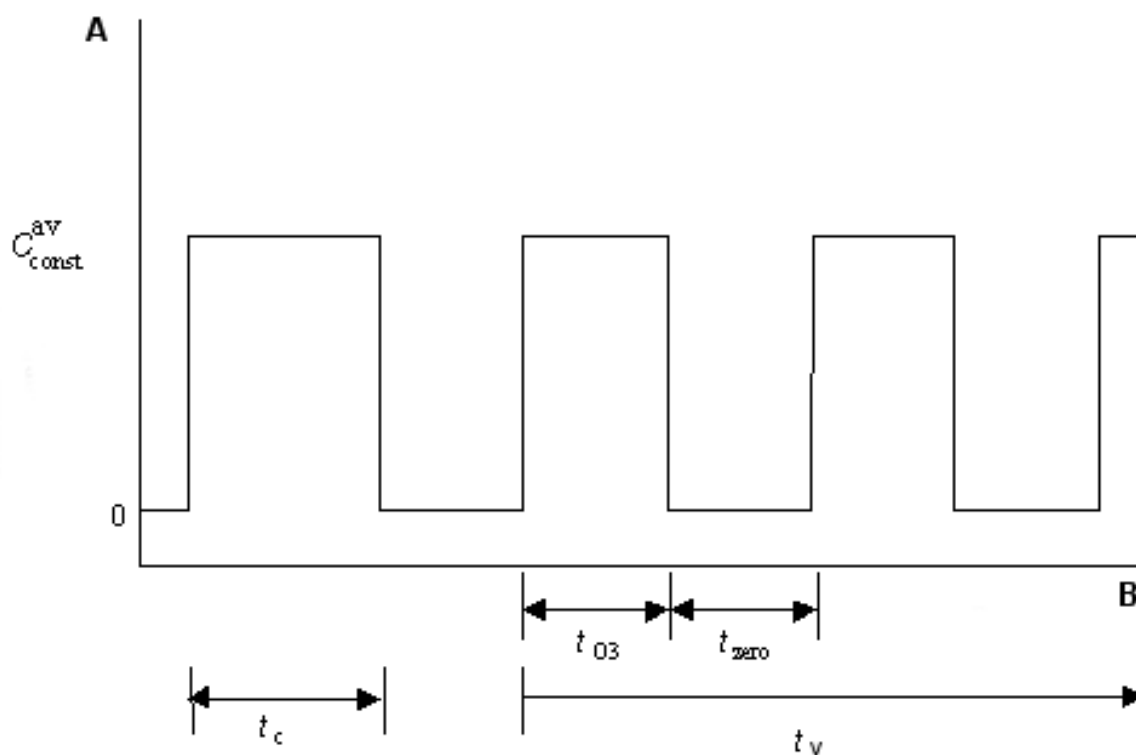
$$X_{av} = \frac{C_{const}^{rav} - 2C_{var}^{rav}}{C_{const}^{rav}} \times 100\% \quad (11)$$

where

X_{av} is the averaging effect (%);

C_{const}^{rav} is the average of at least four independent measurements during the constant concentration period (t_c) (nmol/mol);

C_{var}^{rav} is the average of at least four independent measurements during the variable concentration period (t_v) (nmol/mol).



Key

A Concentration (nmol/mol)

B Time

Figure 2 — Concentration variations for the averaging effect test

8.4.13 Difference sample/calibration port

If the analyser has different ports for feeding sample gas and calibration gas the difference in response of the analyser to feeding through the sample or calibration port shall be tested. The test shall be carried out by feeding the analyser with a test gas with a concentration of 70 % to 80 % of the maximum of the certification range through the sample port. The test shall consist of one independent followed by two individual measurements. After a period of at least four response times the test shall be repeated using the calibration port. The difference shall be calculated according to:

$$D_{sc} = \frac{|x_s - x_c|}{c_t} \times 100 \% \quad (12)$$

where

- D_{sc} is the difference sample/calibration port (%);
- x_s is the average of the measured concentration using the sample port (nmol/mol);
- x_c is the average of the measured concentration using the calibration port (nmol/mol);
- c_t is the concentration of the test gas (nmol/mol).

D_{sc} shall comply with the performance criterion in Table 1.

8.5 Determination of the performance characteristics during the field test

8.5.1 General

The determination of the performance characteristics in the field as a part of the type approval test shall be performed by a designated body. The quality of the materials and equipment used in the described test procedures shall be in accordance with the requirements given in this document.

In the field test during a period of three months two analysers are tested for availability (period of unattended operation), reproducibility in the field and long-term drift. The analysers are run in parallel at one and the same sampling point at a selected monitoring station with specific ambient air conditions (8.5.2). Operational requirements are given for the correct determination of the long-term drift and the reproducibility under field conditions (8.5.3).

8.5.2 Selection of a monitoring station

The selection of a monitoring station is based on the following criteria:

Location

- periurban or rural station.

Monitoring station facilities

- sufficient capacity of the sampling manifold;
- enough room to place two analysers with calibration gases and/or calibration facilities;
- surrounding temperature control for the analysers, climate controlled at $20\text{ °C} \pm 4\text{ °C}$ with temperature registration;
- stable electrical voltage.

Other items that could be considered

- presence of telemetry/telephone facilities for remote surveillance of the functioning of the equipment;
- accessibility.

8.5.3 Operational requirements

After installation of the analysers at the monitoring station the proper functioning of the analysers shall be tested. This comprises (among other things) the proper connections to the sampling manifold, sample gas flows, correct temperatures of e.g. reaction chambers, response to zero and span gases, data transmission and other items, which shall be judged necessary by the designated body.

After verification of the proper functioning, the analysers shall be adjusted at zero and calibrated at a value of about 80 % of the maximum of the certification range.

During the three-month period, the maintenance requirements by the manufacturer of the analyser shall be followed.

Measurements with zero and span gases shall be performed every two weeks. The concentration c_t of the span gas shall be around 90 % of the maximum of the certification range. One independent followed by four individual measurements shall be both performed at zero and at concentration c_t . The measurement results shall be recorded.

To exclude the effect of contamination of the filter when determining the drift of the analyser, the zero and span gases shall be fed to the analyser without passing through the filter.

To avoid the possibility that the filter loading affects the results of the intercomparison of the two analysers and to ensure that the filter loading will not compromise the quality of the air pollution data collected, the filter shall be changed just before each bi-weekly calibration. Filters, which had been preconditioned in the laboratory using O_3 gas mixtures, shall be used.

During the three-month period, no zero and span adjustments shall be made to the analyser, as this will influence the determination of the long-term drift. The measurement data from the analyser shall only be corrected in a mathematical way assuming a linear drift since the last zero and span check.

If an auto-rescaling function or self-correction function is included and considered "normal operational condition", it shall be enabled during the field tests. The magnitude of any self-correction shall be available to the test laboratory. The magnitude of the auto zero and the auto span drift corrections over the period of unattended operation (long-term drift) both have the same restrictions as laid down in the performance characteristics.

For the determination of the various performance characteristics, calibration gases (air containing a certain O_3 concentration) shall be used, traceable to national standards, unless otherwise stated in this document. Various methods for the generation of calibration gases are given in Table 3 (see 8.4.2.3).

The uncertainties in zero and span gases used for the field tests shall proven to be insignificant. Possible contamination of zero and span gas shall not significantly influence the results of the laboratory and field tests. Therefore the calibration gases and zero gas shall meet the following specifications:

Maximum permitted uncertainties in the concentration of gases used for field tests: $\pm 5 \%$.

Specification of the purity of test-gases and zero gas (expressed as absolute value) as given in Table 4.

NOTE It is advised to perform the field tests with the same set of cylinders and zero air generators, exclusively reserved for the tests. The stability of both the zero air and the span gas should be guaranteed over a period longer than the test period.

8.5.4 Long-term drift

After each bi-weekly calibration the drift of the analysers under test shall be calculated at zero and at span following the procedures as given underneath. If the drift compared to the initial calibration exceeds one of the performance criteria for drift at zero or span level, the "period of unattended operation" equals the number of weeks

till the observation of the infringement, minus two weeks. For further (uncertainty) calculations the values for “long term drift” are the values for zero and span drift over the period of unattended operation.

At the beginning of the drift period five individual measurements are recorded (after waiting the time equivalent to one independent measurement just after the calibration) at zero and at span level.

The long-term drift is calculated as follows:

$$D_{L,z} = (C_{z,2} - C_{z,1}) \quad (13)$$

where

$D_{L,z}$ is the drift at zero (nmol/mol);

$C_{z,1}$ is the average concentration of the measurements at zero at the beginning of the drift period (just after the initial calibration) (nmol/mol);

$C_{z,2}$ is the average concentration of the measurements at zero at the end of the drift period (nmol/mol).

$D_{L,z}$ shall comply with the performance criterion in Table 1.

$$D_{L,s} = \frac{(C_{s,2} - C_{s,1}) - D_{L,z}}{C_{s,1}} \times 100 \% \quad (14)$$

where

$D_{L,s}$ is the drift at span concentration c_t as percentage;

$C_{s,1}$ is the average concentration of the measurements at span level at the beginning of the drift period (just after the initial calibration) (nmol/mol);

$C_{s,2}$ is the average concentration of the measurements at span level at the end of the drift period (nmol/mol).

$D_{L,s}$ shall comply with the performance criterion in Table 1.

NOTE For the determination of a systematic or random drift, a graph with the zero and span gas readings can be useful.

8.5.5 Reproducibility standard deviation under field conditions

The reproducibility standard deviation under field conditions is calculated from the measured hourly averaged data during the three-month period.

The difference d_f for each (i th) parallel measurement is calculated from:

$$d_{f,i} = (x_{1,f})_i - (x_{2,f})_i \quad (15)$$

where

$d_{f,i}$ is the i th difference in a parallel measurement (nmol/mol);

$(x_{1,f})_i$ is the i th measurement result of analyser 1 (nmol/mol);

$(x_{2,f})_i$ is the i th measurement result of analyser 2 at the same time as the measurement of analyser 1 (nmol/mol).

The reproducibility standard deviation under field conditions ($s_{r,f}$) is calculated according to:

$$s_{r,f} = \frac{\left(\sqrt{\frac{\sum_{i=1}^n d_{f,i}^2}{2n}} \right)}{av} \times 100 \quad (16)$$

where

$s_{r,f}$ is the reproducibility standard deviation under field conditions (%);

n is the number of parallel measurements;

av is the average value during the field test (nmol/mol).

The reproducibility standard deviation under field conditions, $s_{r,f}$, shall comply with the performance criterion in Table 1.

8.5.6 Period of unattended operation

The period of unattended operation is the time period within which the drift is within the performance criterion for long-term drift. If the manufacturer specifies a shorter period for maintenance, then this will be taken as the period of unattended operation. If one of the analysers malfunctions during the field test, then the field test shall be restarted to show whether the malfunction was coincidental or bad design.

NOTE A minimum period of unattended operation is normally recommended to be at least two weeks.

8.5.7 Period of availability of the analyser

The correct operation of the analysers shall be checked at least every 14 days. It is recommended to perform this check every day during the first 14 days. These checks consist of plausibility checks on the measured values, as well as available status signals and other relevant parameters. Time, duration and nature of any malfunctioning shall be logged.

The total time period with useable measuring data is the period during the field test during which valid measuring data of the ambient air concentrations are obtained. In this time period the time needed for calibrations, conditioning of sample lines (6.3) and filters (6.4) and maintenance shall not be included.

The availability of the analyser is calculated as:

$$A_a = \frac{t_u}{t_t} \times 100 \% \quad (17)$$

where

A_a is the availability of the analyser (%);

t_u is the total time period with validated measuring data;

t_t is the time period of the field test (three months) minus the time for regular calibration, conditioning and maintenance of the instrumentation.

t_u and t_t shall be expressed in the same units (e.g. hours).

The availability of each analyser shall comply with the criterion in Table 1.

8.6 Expanded uncertainty calculation for type approval

The type approval of the analyser consists of the following steps:

- 1) the value of each individual performance characteristic tested in the laboratory shall fulfil the criterion stated in Table 1 (see 8.2);
- 2) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory tests fulfils the criterion as stated in the Council Directive 2002/3/EC. This criterion is the maximum uncertainty of individual measurements for continuous measurements at the hourly alert threshold value value. The relevant specific performance characteristics and the calculation procedure are given in Annex G;
- 3) the value of each of the individual performance characteristics tested in the field shall fulfil the criterion stated in Table 1 (see 8.2);
- 4) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory and field tests fulfils the criterion as stated in the Council Directive 2002/3/EC. This criterion is the maximum uncertainty of individual measurements for continuous measurements at the hourly alert threshold value value. The relevant specific performance characteristics and the calculation procedure are given in Annex G.

The instrument can be type approved when all 4 requirements are met.

9 Field operation and ongoing quality control

9.1 General

When a type approved analyser has been chosen for a particular measuring task, then the suitability of this analyser shall be evaluated at a specific measuring location. This shall be performed by means of a suitability evaluation described in 9.2.

The analyser shall be installed at a monitoring station in such a way that normal operation of the analyser is not compromised. This implies that the analyser is sheltered and shielded from dust, rain and snow, direct sun radiation, strong temperature fluctuations etc. An enclosure (container or building) with temperature control or air conditioning usually fulfils these requirements. At some locations voltage stabilisers for the power supply may be considered, when voltage fluctuations are expected.

After installation of the analyser at the measuring station the analyser shall be tested for proper operation. This is described in 9.3 (initial installation).

Subsequently, once the analyser at the specific site has been judged to conform with the EU data quality objectives, quality assurance and quality control procedures for ongoing monitoring of nitrogen dioxide and nitrogen monoxide concentrations shall be followed (as described in 9.4) in order to ascertain that the measured data comply with the uncertainty requirements as given in the EU legislation.

9.2 Suitability evaluation

9.2.1 General

When a type approved analyser has been chosen, the suitability of this analyser shall be evaluated for the site-specific conditions at the monitoring site. For example, temperature fluctuations might be such that the type approved analyser does not fulfil the uncertainty requirements under these conditions. Consequently it may be necessary to control the temperature of the air directly surrounding the analyser. Ozonetest gases traceable to national standards shall be used for the correct calibration of the analyser.

9.2.2 Analyser for a monitoring station or task

An expanded uncertainty calculation for the type approved analyser shall be made according to EN-ISO/IEC 14956 in combination with the specific circumstances at the monitoring station or site. For the specific concepts of uncertainty calculations, reference is made to ENV 13005 and to Directive 2002/3/EC. The site-specific circumstances, which shall be evaluated, are given in Table 6.

Table 6 — Site-specific conditions to be evaluated

Parameters	Remarks
Sample pressure variation	The sample gas pressure variation during a whole period of a year shall be estimated.
Sample gas temperature variation	The sample gas temperature variation during a whole period of a year shall be estimated. Sample gas temperature may be controlled by heating or thermostating.
Surrounding air temperature variation	The temperature fluctuation shall be within the range specified in the type approval test. Temperature may be controlled thermostatically.
Voltage variation ^a	The voltage fluctuation shall be within the range in the type approval test. Voltage fluctuations may be controlled by means of voltage stabilizers.
H ₂ O concentration range	The H ₂ O concentration range during a whole period of a year shall be estimated.
toluene concentration range	The toluene concentration range during a whole period of a year shall be estimated.
xylene concentration range	The xylene concentration range during a whole period of a year shall be estimated.
Expanded uncertainty of the calibration gas	The expanded uncertainty of the calibration gas shall be included. This implies the uncertainty of the calibration gas itself as well as the uncertainty of any dilution system (where applicable).
Calibration frequency	The intended calibration frequency shall be used for the calculation of the influence of the drift.
^a For an analyser operating on direct current the type approval test of voltage variation shall be carried out over the range of $\pm 10\%$ of the nominal voltage.	

If the site-specific conditions are outside the conditions for which the analyser is type approved, then the analyser shall be retested under these site-specific conditions and a revised type approval will be issued.

The analyser can only be used in the tested configuration.

If the analyser complies with the requirements laid down in EU legislation, then that particular analyser may be installed and used at that monitoring station.

The calculations shall be documented.

9.3 Initial installation

When the analyser and the sampling system have been set up at the monitoring station proper functioning of the analyser and sampling system shall be checked. The results of these checks shall fulfil the requirements and

limitations as set out by the manufacturer of the analyser as well as the requirements (such as materials used, residence times and so on) given in this document. The compliance with the requirements of the manufacturer and requirements set out in this document shall be documented.

When the concentrations measured by an analyser at a monitoring station are collected by a data-logger/computer system, then the proper functioning of the data collection shall be checked. When the measured data are transmitted to a central computer system, the transmission process shall be checked as well. Checks shall be performed to an extent ensuring that the actual concentrations measured by the analyser are properly recorded in any data collection system.

Subsequently each time parts of the data registration/transmission process are changed, the proper function of the complete process shall be rechecked.

All checks on the proper function of the data collection/transmission system(s) shall be documented.

During the initial installation a lack of fit check shall be performed according to 8.4.6. It is permitted that this test be carried out in the laboratory before installation at the site or after installation at the site.

9.4 Ongoing quality control

9.4.1 General

Quality control is necessary in order to ensure that the uncertainties of the measured values for ozone in ambient air are kept within the stated limits during extended continuous monitoring periods in the field. This requires that maintenance, test and calibration procedures shall be followed which are essential for obtaining accurate and traceable air quality data. In this clause, procedures for maintenance, checks and calibration are given. These procedures are regarded as a minimum necessary for maintaining the required quality level.

The quality of the materials and equipment used in the described test procedures shall be in accordance with the requirements given in this document and shall not significantly influence the results of these procedures.

NOTE It is recommended that the designated body that is performing the ongoing quality control procedures is accredited according to EN ISO/IEC 17025.

9.4.2 Frequency of calibrations, checks and maintenance

The checks and calibrations together with their frequency are summarised in Table 7. Criteria are also given for readjustment, calibration or maintenance of the analyser.

Table 7 — Required frequency of calibration, checks and maintenance

Calibration, checks and maintenance	Clause	Frequency	Action criteria ^e
Calibration of the analyser	9.5	At least every three months and after repair	
Certification of test gases	9.6.1	At least every six months	Zero: $\geq$ detection limit Span: $\geq 5,0$ % from last certified value
Zero and span check ^a	9.6.2	At least every two weeks ^b	Zero: ≥ 5 nmol/mol Span: $\geq 5,0$ % of initial span value
Lack of fit check (to be performed in	9.6.3,	Within 1 year after	lack of fit $> 6,0$ %

Calibration, checks and maintenance	Clause	Frequency	Action criteria ^e
laboratory or in field)	Annex B	installation and after repair; further frequency depending on the result of test	of the measured value
Testing sample manifold			
- influence of pressure drop induced by the manifold pump	9.6.5.1		influence > 1 % of measured value
- sample collection efficiency	9.6.5.2	At least every three years	influence > 2 % of the measured value
Change of particulate filters ^c of the sampling system at the sampling inlet and/or at the analyser inlet	6.4, 9.7.1	At least every three months ^d	Response to span gas passing the filter is ≤ 97 %
Test of the sampling lines	6.3	At least every six months ^d	≥ 2 % sample loss
Changing of (if applicable): drying material and other consumables	6.4 9.7.2	At least every six months ^d	As required
Regular maintenance of components of the analyser	9.7.3	As required by manufacturer	As required
^a Span value: recommended concentration of 70 % to 80 % of the certification range. ^b Recommended every 23 h 25 h. ^c The particulate filter shall be changed periodically depending on the dust loading at the sampling site. During this filter change the filter housing shall be cleaned. Overloading of the particulate filter may change the concentration of ozone. ^d Dependent on site-specific conditions. ^e If infringement of an action criterion occurs, corrective actions shall be taken as soon as possible. An evaluation of the influence of the detected infringement on the measurement data produced before the actual correction of the infringement took place shall be given and taken into account during data validation. To ensure that the data capture criterion is met, data will need to be inspected by a trained operator every working day.			

9.5 Calibration of the analyser

Calibration shall be performed at least every three months at a recommended concentration of 70 % to 80 % of the certification range, to determine analyser response and drift. Calibration every two weeks will give a better indication of drift and analyser performance.

Ozone gases traceable to national standards shall be used for the correct calibration of the analyser. Maximum permitted uncertainty in the concentration of gases used for ongoing quality control is ± 5 % with a level of confidence of 95 %. The zero gas shall give no instrument reading higher than the detection limit.

Calibration gases shall be introduced before the filter to check and if necessary to correct for contamination of the filter.

Calibration gases shall be introduced for at least 10 min before any valid readings are taken from the data acquisition system.

Frequency of test:

At least every three months and after repair.

Action criteria:

If zero or span calibration response have drifted beyond the measurement range in use.

Appropriate action:

Adjust or service the analyzer.

Analysers shall not be adjusted in the field unless strictly necessary. All drifts shall be taken into account when processing the data. Analysers shall only be adjusted if the response to either span or zero gas has drifted beyond the end of the valid measurement range of the analyser.

NOTE This is because some analysers take a very long time to stabilise after adjustment; also some site operatives do not correctly record these adjustments. In both these cases large amounts of data have to be considered invalid.

9.6 Checks

9.6.1 Test gases

For the calibration as well as the zero and span checks of the analyser, several methods are available to generate calibration gases. In Table 3 the various methods are given. For each of the methods the user shall demonstrate that the uncertainty of the calibration gas does not add to the expanded uncertainty budget in such a way that the expanded uncertainty for the method is exceeded.

Gases for span and zero checks shall be verified at least every six months with use of reference gases traceable to national standards. The zero gas shall give no instrument readings higher than the detection limit, and the gas used for daily span checks shall not differ by more than 5 % of the last certified value.

The purity of the gases shall be as specified in Table 4.

Action criteria:

Zero $\geq$ detection limit;

Span $\geq$ 5 % of previous verification.

Appropriate action:

Replace the zero-gas cylinder and/or recalibrate the ozone generator.

9.6.2 Zero and span checks

Zero and span gas can be supplied by gas cylinder, or generated by an external calibrator unit or internally in the analyser.

Zero and span gas should be introduced into the analyser for a period of 10 min per gas. The timing of this injection shall coincide with a discrete 10-minutes average collection by the data acquisition system. The timing should also be arranged so that zero gas and span gas are injected during consecutive hourly periods. This timing regime ensures that each hour still contains 75 % of valid data and therefore these hours do not reduce the final data capture.

The differences between two zero or two span values obtained from the following equations should be calculated to determine if the action criteria have been exceeded:

$$X_z = |Z_i - Z_0| \quad (18)$$

where

X_z is the difference between the readings of the recent zero check and the most recent calibration;

Z_i is the reading of the recent zero check;

Z_0 is the reading of the most recent calibration.

$$X_s = \frac{|S_i - S_0|}{S_0} \times 100\% \quad (19)$$

where

X_s is the difference between the readings of the recent span check and the most recent calibration (%);

S_i is the reading of the recent span check;

S_0 is the reading of the most recent calibration.

Frequency of test:

At least every two weeks. Recommended every 23 h or 25 h.

Action criteria:

Zero ≥ 5 nmol/mol of the initial zero span;

Span ≥ 5 % of initial span value.

Appropriate action:

The site should be visited and the correct function of the analyser determined by calibration.

9.6.3 Lack of fit

Delivering a test gas in a field environment, that meets the specifications of the laboratory test, is difficult. As this parameter has already been tested in the laboratory and included in the uncertainty budget, it is not necessary to test this criterion to such a tight specification as for the laboratory tests. It is acceptable to test the lack of fit to a lesser specification (e.g. ± 6 % measured value). If the analyser fails this specification, then the analyser should be removed from site and further, more rigorous testing should occur.

The lack of fit of the analyser shall be tested using the following concentrations: 0 %, 20 %, 60 % and 95 % of the maximum of the certification range of O_3 . At each concentration (including zero) at least two individual readings shall be performed. After each change in concentration at least four response times shall be taken into account before the next measurement is performed.

Frequency of test:

After installation and after repair the analyser shall be tested within 1 year; if the result of this test is $\leq 2,0$ % of the measured value the next test shall be carried out within 3 years; if the result is $> 2,0$ % but $\leq 6,0$ % the analyser shall be retested within 1 year; if the result is $> 6,0$ % appropriate action shall be taken.

Action criteria:

$> 6,0$ % of the measured value.

Appropriate action:

Remove analyser from site for further testing and repair if necessary.

NOTE Lack of fit may be checked either in the laboratory or on site.

9.6.4 Testing the sample manifold

9.6.4.1 Procedure for measuring pressure drop induced by the manifold pump

Equipment required: Handheld manometer.

- 1) Zero manometer with both ports open to atmospheric pressure;
- 2) Disconnect analyser sample pipework from sample manifold;
- 3) Attach –ve port of manometer to sample port on manifold, leave +ve port open to atmospheric pressure;
- 4) Record measured pressure drop in Pa/mbar;
- 5) Disconnect manometer and reconnect analyser sample pipework.

The resulting pressure drop should be used to calculate the induced effect on the analyser's response using the following equation:

$$\Delta R_a = b_{gp} \times \Delta P_m \quad (20)$$

where

- ΔR_a is the change in the analyser's response due to the influence of the pressure drop induced by the manifold pump, expressed as a percentage;
- b_{gp} is the sensitivity coefficient of the analyser to sample gas pressure change expressed as a percentage of the measured value, obtained during the laboratory type approval test;
- ΔP_m is the measured pressure drop induced by the manifold pump.

Action criteria:

Influence of the pressure drop induced by the manifold sampling pump on the measured concentration < 1 %.

Appropriate action:

Reduce flow through manifold to reduce the induced pressure drop so that the criterion in 6.6 is met.

9.6.4.2 Procedure for testing the sample collection efficiency of the sampling system

The flow rate of the test gas in the sampling system should be such that the residence time is greater than or equal to that found under normal operating conditions. Typical manifold systems (diameter ~30 mm, length 2 m) have a volume of ~2 l and shall have a maximum residence time of 5 s.

Frequency of test:

At least every three years

Action criteria:

Sample loss > 2 %.

Appropriate action:

Clean/replace/repair manifold as necessary and re-test.

NOTE A possible test procedure for the sample collection efficiency is given below.

Test gas for O₃ analysers is provided by either dynamic dilution of single or multicomponent cylinders, or from gas cylinders without dilution. Zero-grade air, or better, should be used as a diluent/balance gas for these cylinders.

The results from the tests are in the form of ratio measurement and therefore the correct calibration of the analysers is not necessary, nor is the exact concentration of the test gas. The concentration of the test gas shall, however, be stable.

During testing, the analyser output should be collected through the data collection system at the monitoring site and the normal site operating procedures followed.

Testing comprises two measurements of the test gas, direct sampling of test gas and the sampling of test gas from the complete sampling system. Data averaged over period of 10 min is recorded for each stage of the test.

An example is given in Annex F of possible manifold testing equipment.

Sample system collection efficiency, E_{ss} , is then calculated as follows:

$$E_{ss} = \left(1 - \left(\frac{R_d - R_m}{R_d}\right)\right) \times 100 \% \quad (21)$$

where

E_{ss} is the sample system collection efficiency;

R_d is the mean analyser response to the test gas directly sampled by the analyser;

R_m is the mean analyser response to the test gas via the sample manifold.

9.7 Maintenance

9.7.1 Change of particulate filters

It is suggested that the effective filter life is determined at the original site installation stage and then calibrations made before a filter is changed. New filters should be conditioned with ambient air for 30 min before any ambient data is considered valid. This conditioning time should not be included in any calculation of data capture.

Frequency of test:

Determined from initial site installation, strongly dependent on site conditions.

Action criteria:

Response to span gas passing the filter is $\leq 97 \%$.

Appropriate action:

Calibrate instruments with site standard before changing filter. New filters should be conditioned with ambient air for 30 min before any ambient data is considered valid.

9.7.2 Change of consumables as applicable

The life of all analyser consumables should be determined at the initial installation. Site-specific maintenance periods should be devised for the replacement of such consumables.

9.7.3 Regular maintenance of components of the analyser

The manufacturers recommendations should be followed for the routine maintenance of the analyser.

9.8 Data handling and data reports

The designated body responsible for QA/QC of the monitoring station has the responsibility to produce valid data. This implies that the collected data shall be free from faulty data, zero and span checks or calibrations. When negative values are reported these values shall be included in the calculation of averages. When a value that is larger than the maximum of the certification range is reported, then this value shall be used in the calculation of averages and the said average shall be flagged in such a way that it is clear in the data report that this average might have exceeded the combined uncertainty requirement.

Data collection from the analyser shall be done at a frequency of at least twice per response time period.

Data capture shall be $\geq 75 \%$ of the averaging time.

The response of the analyser to zero or span test gases should be recorded before and after the adjustment of the analyser. Measured drift in the analyser's response that is less than the action criteria in Table 8 shall be corrected for in data processing and not by physical adjustment of the analyser.

The uncertainty of measurement shall be determined and reported every year during operation, e.g. yearly verification or evaluation of data of QA/QC measures. When doing this, as far as possible, the actual values of the performance characteristics obtained during the type approval test shall be used. The specifications of ENV 13005 and EN –ISO 14956 shall be used.

10 Expression of results

The readings from the analyser are converted to concentrations using the appropriate conversion factors and the results expressed in micrograms per cubic metre.

The conversion factors at 20 °C and 101,3 kPa are:

$$1 \mu\text{g O}_3/\text{m}^3 = 0,500 \text{ nmol/mol (V/V) O}_3$$

$$1 \text{ nmol/mol (V/V) O}_3 = 2,00 \mu\text{g O}_3/\text{m}^3$$

11 Test reports and documentation

11.1 Type approval tests

The designated body shall prepare a type approval report, which contains at least the following information:

- complete identification of each specific type of analyser of a series, including software version;
- specification of the type approval tests executed e.g. by reference to this document;
- analyser settings during type approval test;
- series of observations obtained in the type approval tests;
- values determined for the performance characteristics evaluated and the applied evaluation methods (e.g. by reference to this document);
- calculation of the expanded uncertainty of hourly averages.

The type approval report shall be available to the public.

11.2 Field operation

11.2.1 Suitability evaluation

The user and/or operator of an analyser or monitoring station shall prepare a report of the suitability evaluation of the analyser at the monitoring site. The suitability test report shall contain at least the following information:

- reference to this document;
- complete identification of the analyser and monitoring site;
- results of the suitability evaluation of the analyser at the monitoring station;
- proof of the compliance of the sampling system to this document;
- checks on the initial installation.

11.2.2 Documentation

The user and/or operator shall document all maintenance, repairs, calibrations, change of calibration gases, malfunctioning, etc., for each individual analyser, sampling system and monitoring site.

11.2.3 Ambient air quality data reports

Ambient air data reports for ozone shall be prepared according to local, national or EU regulations [3]. The report shall contain at least the following information:

- a) reference to this document;
- b) percentage of data capture;
- c) air quality data presented in the required format
- d) a statement on measurement uncertainty of the data reported.

Annex A (normative)

Correction for ambient nitric oxide

To correct for the effect of reaction of ambient ozone with nitric oxide in the sampling line, the ambient ozone concentration at the inlet of the sampling line (initial ozone concentration) can be calculated according to the following equation:

$$[O_3]_0 = \frac{b \times [O_3]_t}{[O_3]_t - [NO]_t \times e^{(b \times k \times t)}} \quad (A.1)$$

where

$[O_3]_0$ is the ozone concentration at the sampling inlet;

$[O_3]_t$ is the ozone concentration after t seconds of residence time in the sampling line;

$[NO]_t$ is the nitric oxide concentration after t seconds of residence time in the sampling line;

b is the concentration difference between $[O_3]_t$ and $[NO]_t$: $b = [O_3]_t - [NO]_t$ with $b \neq 0$;

k is the rate constant for the reaction of O_3 with NO : $k = 4,43 \times 10^{-4} \text{ nmol/mol}^{-1} \text{ s}^{-1}$ at 25°C ;

t is the residence time in seconds.

Example

Taking the following values:

$[O_3]$: 20 nmol/mol

$[NO]$: 15 nmol/mol

t : 5 s

k : $4,43 \times 10^{-4} \text{ nmol/mol}^{-1} \text{ s}^{-1}$

b : +5 nmol/mol

After applying the formula this results in a initial ozone concentration of $C = 20,7 \text{ nmol/mol}$. The loss of ozone at these concentrations of O_3 and NO is $(20,7 - 20) / 20,7 \times 100 \% = 3,3 \%$

Annex B (normative)

Calculation of lack of fit

B.1 Establishment of the regression line

A linear regression function in the form of $Y_i = A + B \times X_i$ is made through calculation of the function:

$$Y_i = a + B (X_i - X_z). \quad (\text{B.1})$$

For the regression all measuring points (including zero) are taken into account. The total number of measuring points (n) is equal to the number of concentration levels (at least six including zero) times the number of repetitions (at least five) at a particular concentration level.

The coefficient a is obtained from:

$$a = \sum Y_i / n \quad (\text{B.2})$$

where

a is the average value of the Y-values;

Y_i is the individual Y-value;

n is the number of calibration points.

The coefficient B is obtained from:

$$B = \left(\sum Y_i (X_i - X_z) \right) / \sum (X_i - X_z)^2 \quad (\text{B.3})$$

where

X_z is the average of the X-values ($= \sum X_i / n$);

X_i is the individual X-value.

The function $Y_i = a + B (X_i - X_z)$ is converted to $Y_i = A + B \times X_i$ through the calculation of A :

$$A = a - B \times X_z \quad (\text{B.4})$$

B.2 Calculation of the residuals of the averages

The residuals of the averages of each calibration point (including the zero point) are calculated as follows:

The average of each calibration point (including the zero point) at one and the same concentration c is calculated according to:

$$(Y_a)_c = \sum (Y_i)_c / m \quad (\text{B.5})$$

where

$(Y_a)_c$ is the average Y -value at concentration level c ;

$(Y_i)_c$ is the individual Y -value at concentration level c ;

m is the number of repetitions at one and the same concentration level $c = \sum (Y_i/m)_c$.

The residual of each average (d_c) at each concentration level c is calculated according to:

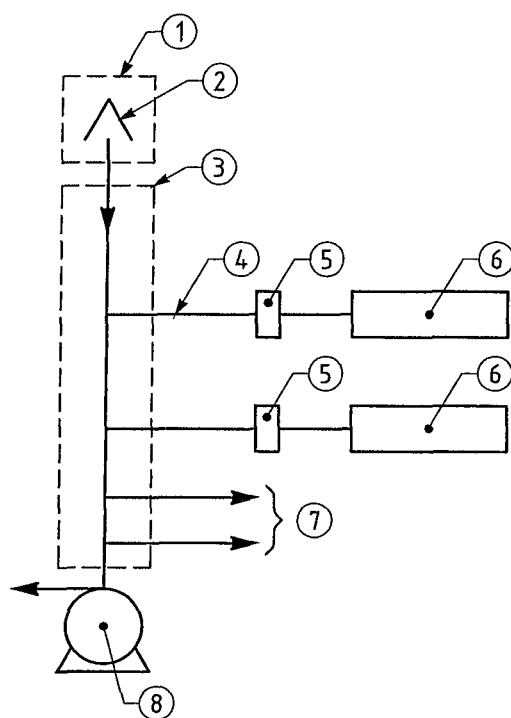
$$d_c = (Y_a)_c - (A + B \times c) \quad (\text{B.6})$$

Each residual to a value relative to its own concentration level c is expressed as:

$$(d_r)_c = \frac{d_c}{c} \times 100 \% \quad (\text{B.7})$$

Annex C (informative)

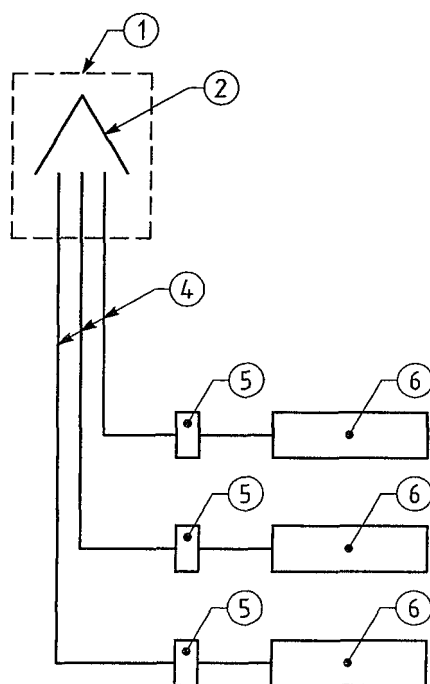
Sampling equipment



Key

- | | |
|---|---|
| 1 | Sample inlet |
| 2 | Rain shield |
| 3 | Manifold |
| 4 | Sampling line |
| 5 | Filter |
| 6 | Analyser |
| 7 | Connection for other analysers or equipment |
| 8 | Sampling pump for the manifold |

Figure C.1 — Sampling layout with a main manifold



Key

- 1 Sample inlet
- 2 Rain shield
- 4 Sampling line
- 5 Filter
- 6 Analyser

Figure C.2 — Sampling layout with individual sampling manifold sampling lines

Annex D (informative)

Sampling on micro scale

The following guidelines should be met as far as practicable:

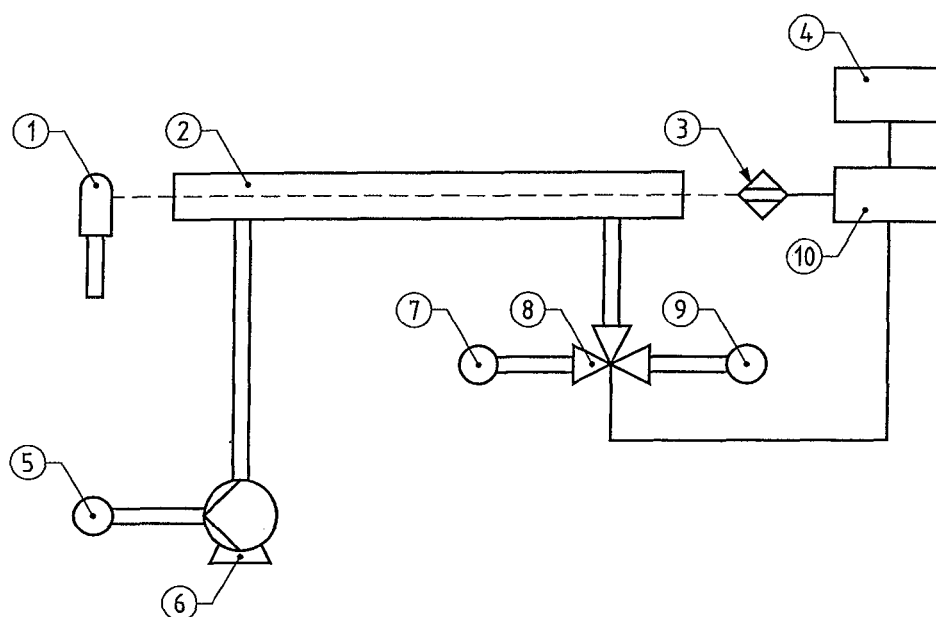
- the flow around the inlet sampling probe should be unrestricted without any obstructions affecting the air flow in the vicinity of the sampler (normally some metres away from buildings, balconies, trees and other obstacles and at least 0,5 m away from the nearest building in the case of sampling points representing air quality at the building line);
- in general, the inlet sampling points should be between 1,5 m (the breathing zone) and 4 m above the ground. Higher positions (up to 8 m) may be necessary in some circumstances. Higher siting may also be appropriate if the station is representative of a large area;
- the inlet probe should not be positioned in the immediate vicinity of sources in order to avoid the direct intake of emissions unmixed with ambient air;
- the sampler's exhaust outlet should be positioned so that recirculation of exhaust air to the sampler inlet is avoided;
- location of traffic-orientated samplers:
 - for all pollutants, such sampling points should be at least 25 m from the edge of major junctions and at least 4 m from the centre of the nearest traffic lane;
 - for nitrogen dioxide, inlets should be no more than 5 m from the kerbside;
 - for particulate matter and lead, inlets should be sited so as to be representative of air quality near to the building line.

The following factors may also be taken into account:

- interfering sources;
- security;
- access;
- availability of electrical power and telephone communications;
- visibility of the site in relation to its surroundings;
- safety of public and operators;
- the desirability of co-locating sampling points for different pollutants;
- planning requirements.

Annex E (informative)

Ultra violet photometric analyser



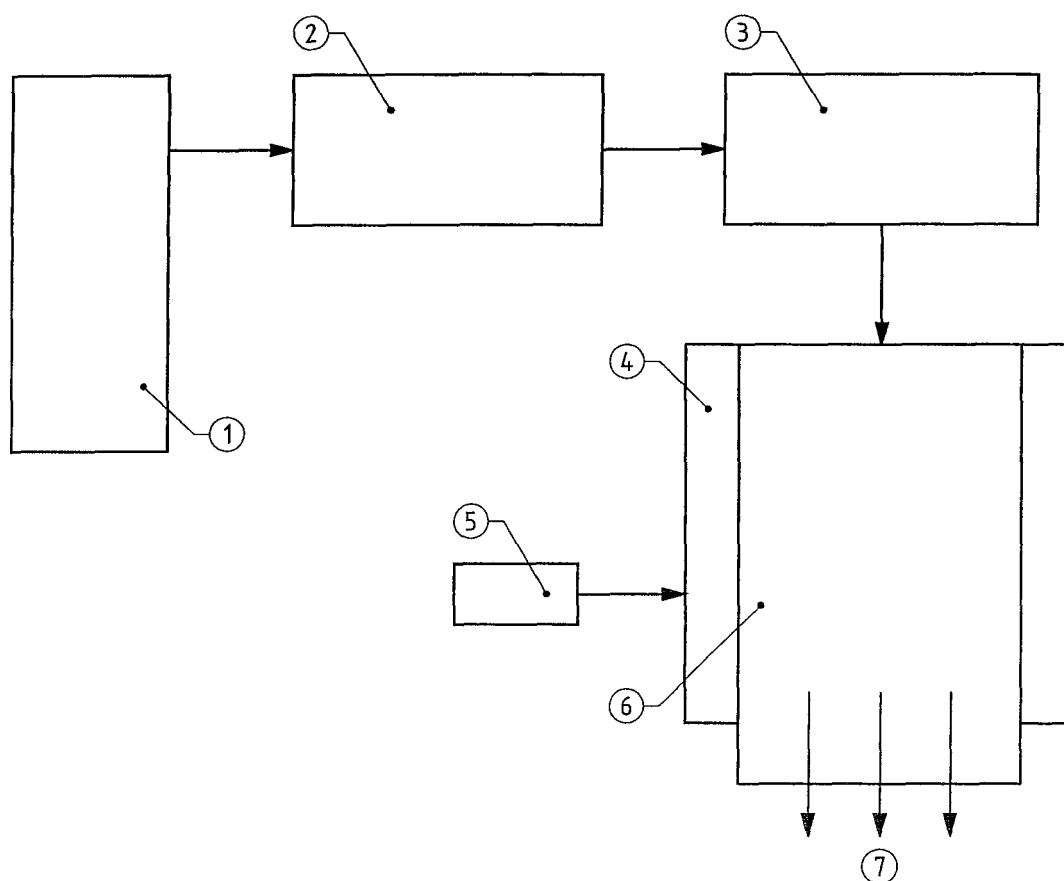
Key

- | | |
|----|---------------------|
| 1 | Source 254 mm |
| 2 | Absorption cell |
| 3 | Detector |
| 4 | Display |
| 5 | Exhaust |
| 6 | Pump |
| 7 | Reference in |
| 8 | Solenoid |
| 9 | Sample in |
| 10 | Digital electronics |

Figure E.1 — Schematic diagram of a dual cell UV ozone analyser

Annex F (informative)

Manifold testing equipment



Key

- 1 Gas cylinder, 200 nmol/mol NO₂, 150 nmol/mol SO₂, 5 nmol/mol CO, balanced air
- 2 50 l/min mass flow controller
- 3 Ozone generator (producing 100 nmol/mol at 50 l/min)
- 4 Tedlar bag, secured around manifold
- 5 Pressure relief valve
- 6 Air sampling manifold
- 7 Analysers

Figure F.1 – Schematic Manifold testing equipment

Annex G (normative)

Type approval

G.1 Type approval and uncertainty calculation

G.1.1 Type approval

The type approval of the analyser consists of the following steps (see Clause 8):

- 1) the value of each individual performance characteristic tested in the laboratory shall fulfil the criterion stated in Table 1 (see 8.2);
- 2) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory tests fulfils the criterion as stated in the Directive 2002/3/EC. This criterion is the maximum uncertainty of hourly values of continuous measurements at the hourly alert threshold value. The relevant specific performance characteristics and the calculation procedure are given in this annex;
- 3) the value of each of the individual performance characteristics tested in the field shall fulfil the criterion stated in Table 1 (see 8.2);
- 4) the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory and field tests fulfils the criterion as stated in the Directive 2002/3/EC. This criterion is the maximum uncertainty of hourly values of continuous measurements at the hourly alert threshold value. The relevant specific performance characteristics and the calculation procedure are given in this annex.

The instrument is type approved when all 4 requirements are met.

G.1.2 Uncertainty calculation

The model equation for the measurement of O_3 is based on the assumption that the measured value c consists of signal contribution from the concentration of O_3 , c' , and a sum of signal contributions c_K (in units of the measured value) caused by the influence of the considered performance characteristics K of the measuring system:

$$c = c' + \sum_K c_K \quad (G.1)$$

From this model equation the following variance equation is derived which describes the uncertainty of the measured value, $u(c)$, under the conditions of the type approval test:

$$u^2(c) = \sum_K u^2(c_K) \quad (G.2)$$

Equation (G.5) is used for estimating the combined standard uncertainty $u(c)$ in type approval step 2. For type approval step 4, which includes the results of the field tests, Equation (G.33) is used for estimating the combined standard uncertainty $u(c)$.

G.2 Type approval requirement 1

Table G.1 gives the performance characteristics that shall be considered in demonstrating compliance with requirement 1 in the type approval procedure and an example of laboratory results.

Table G.1 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Performance criterion for O ₃	Results lab. Test ^b	^c
1	Repeatability standard deviation at zero	$s_{r,z}$	8.4.5	≤ 1,0 nmol/mol	0,45	✓
2	Repeatability standard deviation at concentration c_t (at a level of the hourly alert threshold value)	$s_{r,l}$	8.4.5	≤ 3,0 nmol/mol	2,4	✓
3	Lack of fit (residual from the linear regression function at concentrations higher than zero)		8.4.6			
3a	Largest residual from the linear regression function	X_l		≤ 4,0 % of the measured value	2,5	✓
3a	Residual at zero	$X_{l,z}$	8.4.6	≤ 5,0 nmol/mol	1,4	✓
4	Sensitivity coefficient of sample gas pressure	b_{gp}	8.4.7	≤ 2,0 nmol/mol/kPa	0,20	✓
5	Sensitivity coefficient of sample gas temperature	b_{gt}	8.4.8	≤ 1,0 nmol/molK	0,35	✓
6	Sensitivity coefficient of surrounding temperature	b_{st}	8.4.9	≤ 1,0 nmol/molK	0,20	✓
7	Sensitivity coefficient of electrical voltage	b_v	8.4.10	≤ 0,30 nmol/mol/V	0,25	✓
8	Interferents at zero and concentration c_t (at a level of the hourly alert threshold)	$X_{H_2O,z}$	8.4.11			
8a	H ₂ O with concentration 19 mmol/mol ^d	$X_{H_2O,ct}$		≤ 10 nmol/mol	-2,3	✓
		$X_{H_2S,z}$		≤ 10 nmol/mol	-2,9	✓
8b	toluene with concentration 0,5 µmol/mol	$X_{tol,ct}$		≤ 5,0 nmol/mol	0,49	✓
		$X_{tol,z}$		≤ 5,0 nmol/mol	0,57	✓
8c	xylene with concentration 0,5 µmol/mol	$X_{xyl,ct}$		≤ 5,0 nmol/mol	0,39	✓
		$X_{xyl,z}$		≤ 5,0 nmol/mol	0,47	✓
9	Averaging effect	X_{av}	8.4.12	≤ 7,0 % of the measured value	5,0	✓
13	Short term drift at zero	$D_{l,z}$	8.4.4	≤ 2,0 nmol/mol over 12 h	0,88	✓
14	Short term drift at span level ^a	$D_{l,s}$	8.4.4	≤ 6,0 nmol/mol over 12 h	3,0	✓
15	Response time (rise)	t_r	8.4.3	≤ 180 s	55	✓
16	Response time (fall)	t	8.4.3	≤ 180 s	50	✓

No.	Performance characteristic	Symbol	Clause	Performance criterion for O ₃	Results lab. Test ^b	^c
17	Difference rise time and fall time	t_d	8.4.3	≤ 10 % relative difference or 10 s whatever is the greatest	9,1 %	✓
18	Difference sample/calibration port ^e	D_{sc}	8.4.13	≤ 1,0 %	0,70	✓
^a span level is 70 % to 80 % of the certification range ^b to demonstrate compliance the absolute value of the performance characteristic shall be taken ^c ✓ : requirement is met ^d a H ₂ O-concentration of 19 mmol/mol is about 80 % RH at 20 °C and 101,3 kPa ^e if relevant						

CONCLUSION: All values of the performance characteristics obtained in the laboratory tests are complying with the requirements.

Requirement 1 is fulfilled.

G.3 Type approval requirement 2

In type approval requirement 2 the expanded uncertainty calculated from the standard uncertainties due to the values of the specific performance characteristics obtained in the laboratory tests shall fulfil the criterion as stated in the Council Directive [3]. For this criterion is taken the maximum uncertainty of hourly values of continuous measurements at the hourly alert threshold value.

The values of the following uncertainties shall be included in the calculation of the expanded uncertainty after the laboratory tests.

Table G.2 — Standard uncertainties to be incorporated in the calculation of the expanded uncertainty after the laboratory tests

No.	Standard uncertainty due to	Symbol	Equation, G.
1	Repeatability standard deviation at zero	$u_{r,z}$	6, 7
2	Repeatability standard deviation at the hourly alert threshold value	$u_{r,lv}$	8, 9, 10
3	Lack of fit at the hourly alert threshold value	$u_{l,lv}$	11
4	Variation in sample gas pressure at the hourly alert threshold value	u_{gp}	12, 13
5	Variation in sample gas temperature at the hourly alert threshold value	u_{gt}	14, 15
6	Variation in surrounding temperature at the hourly alert threshold value	u_{st}	16, 17
7	Variation in electrical voltage at the hourly alert threshold value	u_v	18, 19
8	Interferences at the hourly alert threshold value		
8a	H ₂ O with concentration 21 mmol/mol	u_{H_2O}	20, 21, 22, 23
8b	toluene with concentration 0,5 µmol/mol	u_{tol}	24, 25
8c	xylene with concentration 0,5 µmol/mol	u_{xyl}	24, 25
9	Averaging error	u_{av}	30
18	Difference sample/calibration port	u_{Dsc}	32

Next to the uncertainties due to these performance characteristics the uncertainty in the concentration of calibration gas and in the calibration itself are incorporated in the uncertainty calculation.

Table G.3 — Standard uncertainty of the calibration gas to be incorporated in the calculation of the expanded uncertainty after the laboratory tests

No.	Standard uncertainty due to	Symbol	Equation, G.
21	Uncertainty in calibration gas	u_{cg}	31

The calculation of standard uncertainties is based on the procedures laid down in EN ISO 14956. The uncertainty calculation shall be carried out with the values of the performance characteristics at the concentration of the hourly alert threshold value (if relevant).

The absolute **expanded uncertainty**, U_c , shall be calculated according to:

$$U_c = k \times u_c \quad (\text{G.3})$$

with $k = 2$

where

U_c is the expanded uncertainty (nmol/mol);

k is the coverage factor of approximately 95 %;

u_c is the combined standard uncertainty (nmol/mol);

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to:

$$U_{c,rel} = (U_c / hlv) \times 100 \% \quad (\text{G.4})$$

where

$U_{c,rel}$ is the relative combined expanded uncertainty (%);

U_c is the combined expanded uncertainty (nmol/mol);

hlv is the hourly alert threshold value (nmol/mol).

Requirement 2 is fulfilled when: $U_{c,rel} \leq U_{req,rel}$.

The **combined standard uncertainty**, u_c , shall be calculated according to:

$$u_c = \sqrt{u_{r,z}^2 + u_{r,lv}^2 + u_{l,lv}^2 + u_{gp}^2 + u_{gt}^2 + u_{st}^2 + u_v^2 + u_{H_2O}^2 + (u_{int,pos}^2 \text{ OR } u_{int,neg}^2) + u_{av}^2 + u_{Dsc}^2 + u_{cg}^2} \quad (\text{G.5})$$

where

u_c is the combined standard uncertainty (nmol/mol);

$u_{r,z}$ is the standard uncertainty for repeatability at zero (nmol/mol);

$u_{r,lv}$ is the standard uncertainty for repeatability at level of the hourly alert threshold value (nmol/mol);

$u_{l,lv}$ is the standard uncertainty for lack of fit at level of the hourly alert threshold value (nmol/mol);

u_{gp} is the standard uncertainty for sample gas pressure variation (nmol/mol);

u_{gt} is the standard uncertainty for sample gas temperature variation (nmol/mol);

u_{st} is the standard uncertainty for surrounding temperature variation (nmol/mol);

u_V is the standard uncertainty for electrical voltage variation (nmol/mol);

u_{H_2O} is the standard uncertainty for the presence of water vapour (nmol/mol);

$u_{int,pos}$ is the standard uncertainty for interferents (except water vapour) with positive impact (nmol/mol);

$u_{int,neg}$ is the standard uncertainty for interferents (except water vapour) with negative impact (nmol/mol);

u_{av} is the standard uncertainty for averaging (nmol/mol);

u_{Dsc} is the standard uncertainty for difference sample/calibration port (nmol/mol);

u_{cg} is the standard uncertainty for calibration gas (nmol/mol).

For the calculation of the standard uncertainties the following equations shall be used.

The standard uncertainty for **repeatability at zero**, $u_{r,z}$, is calculated according to:

$$u_{r,z} = s_{r,z} / \sqrt{m} \quad (G.6)$$

with

$$m = t / ((t_r + t_f) / 2) \quad (G.7)$$

where

$u_{r,z}$ is the standard uncertainty for repeatability at zero (nmol/mol);

t is 3 600 s;

t_r is the response time (rise) (s);

t_f is the response time (fall) (s);

$s_{r,z}$ is the repeatability standard deviation at zero according to 8.4.5 (nmol/mol);

m is the number of individual measurements in 3 600 s, taking response (rise) and response time (fall) for half that time period respectively;

The standard uncertainty for **repeatability at the hourly alert threshold value**, $u_{r,lv}$, is calculated according to:

$$u_{r,lv} = s_{r,lv} / \sqrt{m} \quad (G.8)$$

with

$$m = t / ((t_r + t_f) / 2) \quad (G.9)$$

and

$$s_{r,lv} = (h\nu / c_t) \times s_{r,ct} \quad (G.10)$$

where

- $u_{r,lv}$ is the standard uncertainty for repeatability at the hourly alert threshold value (nmol/mol);
- $s_{r,lv}$ is the repeatability standard deviation at the hourly alert threshold value (nmol/mol);
- m is the number of individual measurements in 3 600 s, taking response (rise) and response time (fall) for half that time period respectively
- t is 3 600 s;
- t_r is the response time (rise) (s);
- t_f is the response time (fall) (s);
- h/v is the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration (at the level of the hourly alert threshold) (nmol/mol);
- $s_{r,ct}$ is the repeatability standard deviation at the test gas concentration (nmol/mol).

The standard uncertainty due to **lack of fit at the hourly alert threshold value**, $u_{l,lv}$, is calculated according to:

$$u_{l,lv} = ((X_{l,lv} / 100) \times h/v) / \sqrt{3} \quad (G.11)$$

where

- $u_{l,lv}$ is the standard uncertainty due to lack of fit at the hourly alert threshold value (nmol/mol);
- $X_{l,lv}$ is the calculated residual from a linear regression function at the hourly alert threshold value (%) (calculated according to Annex B);
- h/v is the hourly alert threshold value (nmol/mol).

The standard uncertainty due to variation of **sample gas pressure at the hourly alert threshold value**, u_{sp} , is calculated according to:

$$u_{sp} = ((h/v/c_t) \times b_{gp}) \times (\Delta_{gp} / \sqrt{3}) \quad (G.12)$$

with

$$\Delta_{gp} = P_1 - P_2 \quad (G.13)$$

where

- u_{gp} is the standard uncertainty due to the influence of pressure (nmol/mol);
- b_{gp} is the sensitivity coefficient of sample gas pressure variation (%/kPa);
- h/v is the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);
- Δ_{gp} is the pressure range, used in the laboratory test (kPa);
- P_1 is the sample gas pressure P_1 (kPa);
- P_2 is the sample gas pressure P_2 (kPa).

The standard uncertainty due to variation of **sample gas temperature at the hourly alert threshold value**, u_{gt} , is calculated according to:

$$u_{gt} = ((hlv / c_t) \times b_{gt}) \times (\Delta_{gt} / \sqrt{3}) \quad (G.14)$$

with

$$\Delta_{gt} = T_{gt,1} - T_{gt,2} \quad (G.15)$$

where

u_{gt} is the standard uncertainty due to the variation of sample gas temperature (nmol/mol);

hlv is the hourly alert threshold value (nmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);

b_{gt} is the sensitivity coefficient of sample gas temperature variation (nmol/mol);

Δ_{gt} is the sample gas temperature range, used in the laboratory test (K);

$T_{gt,1}$ is the sample gas temperature T_1 (in K);

$T_{gt,2}$ is the sample gas temperature T_2 (in K).

The standard uncertainty due to variation of **surrounding temperature at the hourly alert threshold**, u_{st} , is calculated according to:

$$u_{st} = ((hlv / c_t) \times b_{st}) \times (\Delta_{st} / \sqrt{3}) \quad (G.16)$$

with

$$\Delta_{st} = T_{st,1} - T_{st,2} \quad (G.17)$$

where

u_{st} is the standard uncertainty due to the variation of the surrounding temperature (nmol/mol);

hlv is the hourly alert threshold value (nmol/mol);

c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);

b_{st} is the sensitivity coefficient of the surrounding temperature variation (nmol/mol/K);

Δ_{st} is the surrounding temperature range, used in the laboratory test (K);

$T_{st,1}$ is the surrounding temperature T_1 (°C);

$T_{st,2}$ is the surrounding temperature T_2 (°C).

The standard uncertainty due to variation of **electrical voltage at the hourly alert threshold value**, u_v , is calculated according to:

$$u_v = ((hlv / c_t) \times b_v) \times (\Delta_v / \sqrt{3}) \quad (G.18)$$

with

$$\Delta_v = V_1 - V_2 \quad (\text{G.19})$$

where

- u_v is the standard uncertainty due to the variation of electrical voltage (nmol/mol);
- h/v is the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);
- b_v is the sensitivity coefficient of electrical voltage variation (nmol/mol/V);
- Δ_v is the electrical voltage range, used in the laboratory test (V);
- V_1 is the electrical voltage V_1 (V);
- V_2 is the electrical voltage V_2 (V).

The influence quantity of water vapour is established at a water concentration of 19 mmol/mol. The uncertainty, however, is to be established at a water concentration of 21 mmol/mol. The standard uncertainty due to **interference by the presence of water vapour at the hourly alert threshold value**, $u_{\text{H}_2\text{O}}$, is therefore calculated according to:

$$X_{\text{H}_2\text{O},z,\max} = (18/16) \times X_{\text{H}_2\text{O},z} \quad (\text{G.20})$$

$$X_{\text{H}_2\text{O},c_t,\max} = (18/16) \times X_{\text{H}_2\text{O},c_t} \quad (\text{G.21})$$

$$X_{\text{H}_2\text{O},\max} = ((X_{\text{H}_2\text{O},c_t,\max} - X_{\text{H}_2\text{O},z,\max}) / c_t) \times h/v + X_{\text{H}_2\text{O},z,\max} \quad (\text{G.22})$$

$$u_{\text{H}_2\text{O}} = |X_{\text{H}_2\text{O}} / c_{\text{H}_2\text{O},\max}| \times \sqrt{(c_{\text{H}_2\text{O},\max}^2 + c_{\text{H}_2\text{O},\max} \times c_{\text{H}_2\text{O},\min} + c_{\text{H}_2\text{O},\min}^2) / 3} \quad (\text{G.23})$$

where

- $X_{\text{H}_2\text{O},z,\max}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at zero concentration of the measurand (nmol/mol);
- $X_{\text{H}_2\text{O},z}$ is the influence quantity of an H_2O concentration of 19 mmol/mol at zero concentration of the measurand (nmol/mol);
- $X_{\text{H}_2\text{O},c_t,\max}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at the test concentration c_t of the measurand (nmol/mol);
- $X_{\text{H}_2\text{O},c_t}$ is the influence quantity of an H_2O concentration of 19 mmol/mol at the test concentration c_t of the measurand (nmol/mol);
- $X_{\text{H}_2\text{O}}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at the hourly alert threshold (nmol/mol);
- c_t is the test gas concentration of the measurand (nmol/mol);
- h/v is the hourly alert threshold value (nmol/mol);

$u_{\text{H}_2\text{O}}$ is the standard uncertainty due to interference by the presence of water vapour (nmol/mol);

$c_{\text{H}_2\text{O,max}}$ is the maximum concentration of water vapour (mmol/mol) (= 21 mmol/mol);

$c_{\text{H}_2\text{O,min}}$ is the minimum concentration of water vapour (mmol/mol) (= 6 mmol/mol).

The standard uncertainty due to each **interfering compound at the hourly alert threshold value** (other than water vapour), u_{int} , is calculated according to:

$$X_{\text{int}} = ((X_{\text{int},c_t} - X_{\text{int},z}) / c_t) \times hlv + X_{\text{int},z} \quad (\text{G.24})$$

$$u_{\text{int}} = |X_{\text{int}} / c_{\text{int,max}}| \times \sqrt{(c_{\text{int,max}}^2 + c_{\text{int,max}} \times c_{\text{int,min}} + c_{\text{int,min}}^2) / 3} \quad (\text{G.25})$$

where

X_{int,c_t} is the influence quantity of the maximum concentration (nmol/mol resp. $\mu\text{mol/mol}$) of the relevant interfering compound at the test concentration c_t of the measurand (nmol/mol);

$X_{\text{int},z}$ is the influence quantity of the maximum concentration (nmol/mol resp. $\mu\text{mol/mol}$) of the relevant interfering compound at zero concentration of the measurand (nmol/mol);

X_{int} is the influence quantity of the relevant interfering compound (nmol/mol);

c_t is the test concentration of the measurand at the level of the hourly alert threshold value (nmol/mol);

hlv is the hourly alert threshold value (nmol/mol);

u_{int} is the standard uncertainty due to interference by the presence of a chemical compound (nmol/mol);

$c_{\text{int,max}}$ is the maximum concentration of interfering compound (nmol/mol resp. $\mu\text{mol/mol}$);

$c_{\text{int,min}}$ is the minimum concentration of interfering compound (nmol/mol resp. $\mu\text{mol/mol}$).

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated.

$$S_{u_{\text{int,pos}}} = u_{\text{int},1,\text{pos}} + u_{\text{int},2,\text{pos}} + \dots + u_{\text{int},n,\text{pos}} \quad (\text{G.26})$$

$$S_{u_{\text{int,neg}}} = u_{\text{int},1,\text{neg}} + u_{\text{int},2,\text{neg}} + \dots + u_{\text{int},n,\text{neg}} \quad (\text{G.27})$$

Take the highest sum as the representative value for all interferents.

$$u_{\text{int,pos}} = \sqrt{(u_{\text{int},1,\text{pos}} + u_{\text{int},2,\text{pos}} + \dots + u_{\text{int},n,\text{pos}})^2} \quad (\text{G.28})$$

$$u_{\text{int,neg}} = \sqrt{(u_{\text{int},1,\text{neg}} + u_{\text{int},2,\text{neg}} + \dots + u_{\text{int},n,\text{neg}})^2} \quad (\text{G.29})$$

where

$u_{\text{int,pos}}$ is the sum of uncertainties due to interferents with positive impact (nmol/mol);

$u_{\text{int},1,\text{pos}}$ is the uncertainty due to the 1th interferent with positive impact (nmol/mol);

$u_{\text{int},n,\text{pos}}$ is the uncertainty due to the n th interferent with positive impact (nmol/mol);

$u_{\text{int},\text{neg}}$ is the sum of uncertainties due to interferents with negative impact (nmol/mol);

$u_{\text{int},1,\text{neg}}$ is the uncertainty due to the 1th interferent with negative impact (nmol/mol);

$u_{\text{int},n,\text{neg}}$ is the uncertainty due to the n th interferent with negative impact (nmol/mol).

The standard uncertainty due to the **averaging error**, u_{av} , is calculated according to:

$$u_{\text{av}} = ((X_{\text{av}} / 100) \times hlv) / \sqrt{3} \quad (\text{G.30})$$

where

u_{av} is the standard uncertainty due to the averaging error (nmol/mol);

X_{av} is the averaging error (% of the measured value);

hlv is the hourly alert threshold value (nmol/mol).

The standard uncertainty due to the **calibration gas**, u_{cg} , is calculated according to:

$$u_{\text{cg}} = ((X_{\text{cg}} / 100) \times hlv) / 2 \quad (\text{G.31})$$

where

u_{cg} is the standard uncertainty due to the calibration gas (nmol/mol);

X_{cg} is the expanded uncertainty of the calibration gas (%);

hlv is the hourly alert threshold value (nmol/mol).

The standard uncertainty due to the **difference sample/calibration port**, u_{Dsc} , is calculated according to:

$$u_{\text{Dsc}} = ((D_{\text{sc}} / 100) \times hlv) / \sqrt{3} \quad (\text{G.32})$$

where

u_{Dsc} is the standard uncertainty due to the difference sample/calibration port (nmol/mol);

D_{sc} is the difference sample/calibration port (%);

hlv is the hourly alert threshold value (nmol/mol).

G.4 Example Result of type approval for O₃ with the results of the laboratory tests

G.4.1 Values of test parameters to be used in the uncertainty calculations

Table G.4 — Physical influences

	test range	lower value	upper value
Sample pressure variation	30 kPa	80 kPa	110 kPa
Sample gas temperature variation	30 K	0 °C	30 °C
Surrounding air temperature variation ^a	30 K	0 °C	30 °C
Voltage variation ^a	40 V	210 V	250 V
^a specified by the manufacturer			

Table G.5 — Chemical influences/interferents

Interferents		lower value, $c_{\text{int,min}}$	upper value, $c_{\text{int,max}}$
H ₂ O	mmol/mol	6	21
toluene	μmol/mol	0	0,5
xylene	μmol/mol	0	0,5

Table G.6 — Test concentrations of the measurand.

c_t at a level of the hourly alert threshold value	120	nmol/mol
c_t at span level (70 % to 80 % of the certification range)	190	nmol/mol

Table G.7 — Default values

Uncertainty in calibration gas concentration, expressed at a level of confidence of 95 %	± 3 %
--	-------

G.4.2 Uncertainty due to interferents, except water vapour

First, calculate the summed uncertainties due to the interferents with positive impact and the summed uncertainties due to the interferents with negative impact. Take the highest sum as the representative value for all interferents.

Table G.8 — Standard uncertainty due to interferents, except water vapour

nr	Standard uncertainty due to	Result		Equation G.	uncert. $u(p)$ nmol/mol	uncert. $u(p)^2$ (nmol/mol) ²
8	Interferents					
8b	toluene with concentration 0,5 $\mu\text{mol/mol}$ at zero	0,49	nmol/mol O ₃			
	toluene with concentration 0,5 $\mu\text{mol/mol}$ at c_t	0,57	nmol/mol O ₃			
	benzene with concentration 0,5 $\mu\text{mol/mol}$ at the hourly alert threshold value	0,57	nmol/mol O ₃	24, 25	0,33	0,11
8c	xylene with concentration 0,5 $\mu\text{mol/mol}$ at zero	0,39	nmol/mol O ₃			
	xylene with concentration 0,5 $\mu\text{mol/mol}$ at c_t	0,47	nmol/mol O ₃			
	xylene with concentration 0,5 $\mu\text{mol/mol}$ at the hourly alert threshold value	0,47	nmol/mol O ₃	24, 25	0,27	0,07
8b-8c	summed uncertainty with positive impact			26, 27, 28, 29	0,60	
8b-8c	summed uncertainty with negative impact			26, 27, 28, 29	0,00	

CONCLUSION: Take the summed uncertainty with positive impact for further calculation.

G.4.3 Calculation of expanded uncertainty from the results of laboratory tests for O₃

The following test values have been used for calculating the uncertainties.

Chemical influences/interferents

Table G.9 – Chemical influences/interferents

		lower value, $c_{\text{H}_2\text{O},\text{min}}$	upper value, $c_{\text{H}_2\text{O},\text{max}}$
H ₂ O	mmol/mol	6	21

The tests are carried out with the following test concentration of the measurand.

Table G.10 – Test concentration of the measurand

$c_t(\text{O}_3)$	nmol/mol	125
-------------------	----------	-----

The test values for sample pressure variation, sample gas temperature variation, surrounding air temperature variation and voltage variation shall be taken. In the calculation in this annex the following values are used in the calculation of the standard uncertainties:

Table G.11 – Values in the calculation of the standard uncertainties

	test range	lower value	upper value
Sample pressure variation	30 kPa	80 kPa	110 kPa
Sample gas temperature variation	30 K	0 °C	30 °C
Surrounding air temperature variation ^a	30 K	0 °C	30 °C
Voltage variation ^a	40 V	210 V	250 V
^a specification of the manufacturer			

Table G.12 — Default values

Uncertainty in calibration gas concentration, expressed as expanded uncertainty	± 3 %
---	-------

Table G.13 — Uncertainty from the results of laboratory tests

No	Standard uncertainty due to	Result or default value		Equation G.	uncert. u(p) nmol/mol	uncert. u(p) ² (nmol/mol) ²
1	Repeatability standard deviation at zero	0,45	nmol/mol	6, 7	0,05	0,00
2	Repeatability standard deviation at level of the hourly alert threshold value	2,4	nmol/mol	8, 9, 10	0,18	0,03
3	Lack of fit at the hourly alert threshold value	2,5	% of the tested value	11	1,73	3,00
4	Sensitivity coefficient of sample gas pressure	0,20	nmol/mol/kPa	12, 13	2,19	4,79
5	Sensitivity coefficient of sample gas temperature	0,35	nmol/mol/K	14, 15	3,83	14,7
6	Sensitivity coefficient of surrounding temperature	0,20	nmol/mol/K	16, 17	2,19	4,79
7	Sensitivity coefficient of electrical voltage	0,25	nmol/mol/V	18, 19	3,65	13,3
8	Interferents at level of the hourly alert threshold value					
	8a H ₂ O with concentration 21 mmol/mol	-3,2 ^a	nmol/mol	20, 21, 22, 23	-2,23	4,99
	8b - 8c summed uncertainty with positive			24, 25, 26, 27, 28, 29	0,60	0,36

No	Standard uncertainty due to	Result or default value		Equation G.	uncert. u(p) nmol/mol	uncert. u(p) ² (nmol/mol) ²
	or negative impact (whichever is the greatest)					
9	Averaging error	5,0	% of the tested value	30	3,46	12,0
18	Difference sample calibration port	0,7	%	32	0,48	0,24
21	Uncertainty in calibration gas	3,0	% of limit value	31	1,80	3,24
^a calculated by extrapolation from the test results at 19 mmol/mol (see Equation G.11)						

Table G.14 — Result of the calculation

combined standard uncertainty	7,84	nmol/mol
expanded uncertainty	15,68	nmol/mol
expanded uncertainty actual	13,1	%
expanded uncertainty required	15	%

CONCLUSION: $U_{c, rel} \leq U_{req, rel}$

Requirement 2 is fulfilled.

G.5 Type approval requirement 3

Table G.14 gives the performance characteristics that shall be considered in demonstrating compliance with requirement 3 in the type approval procedure.

Table G.15 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Performance criterion for O ₃	Results field test	^b
10	Reproducibility standard deviation under field conditions	$S_{r,f}$	8.5.5	$\leq 5,0$ % of the average of a three month period	1,7	✓
11	Long term drift at zero	$D_{l,z}$	8.5.4	$\leq 5,0$ nmol/mol	0,99	✓
12	Long term drift at span level ^a	$D_{l,s}$	8.5.4	$\leq 5,0$ % of maximum of certification range	3,2	✓
19	Period of unattended operation		8.5.6	> 3 months or less if manufacturer indicates a shorter period	4	✓
20	Availability of the analyser	A_a	8.5.7	> 90 %	95	✓
^a span level is 70 % to 80 % of the certification range						
^b ✓ : requirement is met						

CONCLUSION: All values of the performance characteristics obtained in the field tests are complying with the requirements.

Requirement 3 is fulfilled.

G.6 Type approval requirement 4

The values of the following uncertainties shall be included in the calculation of the expanded uncertainty after the laboratory and field tests:

Table G.16 — Standard uncertainties to be incorporated in the calculation of the expanded uncertainty after the laboratory and field tests

No.	Standard uncertainty due to	Symbol	Equation, G.
1	Repeatability at zero	$u_{r,z}$	6, 7
2	Repeatability at the hourly alert threshold value ^a	$u_{r,lv}$	8, 9, 10
3	Lack of fit at the hourly alert threshold value	$u_{l,lv}$	11
4	Variation in sample gas pressure at the hourly alert threshold value	u_{gp}	12, 13
5	Variation in sample gas temperature at the hourly alert threshold value	u_{gt}	14, 15

No.	Standard uncertainty due to	Symbol	Equation, G.
6	Variation in surrounding temperature at the hourly alert threshold value	u_{st}	16, 17
7	Variation in electrical voltage at the hourly alert threshold value	u_v	18, 19
8	Interferents		
8a	H ₂ O with concentration 21 mmol/mol	u_{H_2O}	20, 21, 22, 23
8b	toluene with concentration 0,5 µmol/mol	u_{tol}	24, 25
8c	xylene with concentration 0,5 µmol/mol	u_{xyl}	24, 25
8b	benzene with concentration 1 nmol/mol	u_{benz}	24, 25
9	Averaging error	u_{av}	30
10	Reproducibility under field conditions ^a	$u_{r,f}$	34
11	Long term drift at zero	$u_{d,l,z}$	35
12	Long term drift at the hourly alert threshold valuee	$u_{d,l,lv}$	36
18	Difference sample/calibration port	u_{Dsc}	32
^a Take for the calculation of the combined standard uncertainty the uncertainty due to the repeatability at the hourly alert threshold value or the uncertainty due to the reproducibility under field conditions, whichever is the greatest.			

Next to the uncertainties due to these performance characteristics the uncertainty in the concentration of calibration gas and in the calibration itself are incorporated in the uncertainty calculation.

Table G.17 — Standard uncertainty of the calibration gas to be incorporated in the calculation of the expanded uncertainty after the laboratory and field tests

No.	Standard uncertainty due to	Symbol	Equation, G.
20	Uncertainty in calibration gas	u_{cg}	31

The calculation of standard uncertainties is based on the procedures laid down in EN ISO 14956. The uncertainty calculation shall be carried out with the values of the performance characteristics at the concentration of the hourly alert threshold value (if relevant).

The absolute **expanded uncertainty**, U_c , shall be calculated according to:

$$U_c = k \times u_c \quad (G.3)$$

with $k = 2$

where

U_c is the expanded uncertainty (nmol/mol);

k is the coverage factor of approximately 95 %;

u_c is the combined standard uncertainty (nmol/mol).

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to:

$$U_{c,rel} = (U_c / hlv) \times 100 \% \quad (G.4)$$

where

$U_{c,rel}$ is the relative expanded uncertainty (%);

U_c is the expanded uncertainty (nmol/mol);

hlv is the hourly alert threshold (nmol/mol).

Requirement 4 is fulfilled when: $U_{c,rel} \leq U_{req,rel}$

The **combined standard uncertainty**, u_c , is calculated using the formula:

$$u_c = \sqrt{u_{r,z}^2 + (u_{r,lv}^2 \text{ or } u_{r,f}^2) + u_l^2 + u_{gp}^2 + u_{gt}^2 + u_{st}^2 + u_v^2 + u_{H_2O}^2 + (u_{int,pos}^2 \text{ or } u_{int,neg}^2) + u_{av}^2 + u_{d,l,z}^2 + u_{d,l,lv}^2 + u_{D_{sc}}^2 + u_{cg}^2} \quad (G.33)$$

with

$u_{r,z}$ is the standard uncertainty for repeatability at zero (nmol/mol);

$u_{r,lv}$ is the standard uncertainty for repeatability at the hourly alert threshold value (nmol/mol);

u_l is the standard uncertainty for lack of fit at the hourly alert threshold value (nmol/mol);

u_{gp} is the standard uncertainty for sample gas pressure variation (nmol/mol);

u_{gt} is the standard uncertainty for sample gas temperature variation (nmol/mol);

u_{st} is the standard uncertainty for surrounding temperature variation (nmol/mol);

u_v is the standard uncertainty for electrical voltage variation (nmol/mol);

u_{H_2O} is the standard uncertainty for the presence of water vapour (nmol/mol);

$u_{int,pos}$ is the standard uncertainty for interferents (except water vapour) with positive impact (nmol/mol);

$u_{int,neg}$ is the standard uncertainty for interferents (except water vapour) with negative impact (nmol/mol);

u_{av} is the standard uncertainty for averaging (nmol/mol);

$u_{r,f}$ is the standard uncertainty for reproducibility under field conditions (nmol/mol);

$u_{d,l,z}$ is the standard uncertainty for long term drift at zero (nmol/mol);

$u_{d,l,lv}$ is the standard uncertainty for long term drift at level of the hourly alert threshold value (nmol/mol);

$u_{D_{sc}}$ is the standard uncertainty for difference sample/calibration port (nmol/mol);

u_{cg} is the standard uncertainty for calibration gas (nmol/mol).

For the calculation of the standard uncertainties the test values as given in G.4 have been used.

For the calibration frequency the following value has been used:

Calibration frequency: 1 x / 3 month

The standard uncertainty for **repeatability at zero**, $u_{r,z}$, is calculated according to Equations (G.6, G.7).

The standard uncertainty for **repeatability at the hourly alert threshold value**, $u_{r,hv}$, is calculated according to Equations (G.8, G.9, G.10)

The standard uncertainty due to **lack of fit at the hourly alert threshold value**, u_l , is calculated according to Equation (G.11).

The standard uncertainty due to variation of **sample gas pressure at the hourly alert threshold value**, u_{sp} , is calculated according to Equations (G.12, G.13).

The standard uncertainty due to variation of **sample gas temperature at the hourly alert threshold value**, u_{gt} , is calculated according to Equations (G.14, G.15).

The standard uncertainty due to variation of **surrounding temperature at the hourly alert threshold value**, u_{gt} , is calculated according to Equations (G.16, G.17).

The standard uncertainty due to variation of **electrical voltage at the hourly alert threshold value**, u_v , is calculated according to Equations (G.18, G.19).

The standard uncertainty due to **interference by the presence of water vapour**, u_{H_2O} , is calculated according to Equations (G.20, G.21, G.22, G.23).

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated. Take the highest sum as the representative value for all interferents. See Equations (G.24, G.25), (G.26, G.27) and (G.28, G.29).

The standard uncertainty due to the **averaging error**, u_{av} , is calculated according to Equation (G.30).

The standard uncertainty due to the **reproducibility under field conditions at the hourly alert threshold value**, $u_{r,f}$, is calculated according to:

$$u_{r,f} = hv \times (s_{r,f}/100) \quad (G.34)$$

where

$u_{r,f}$ is the standard uncertainty due to reproducibility under field conditions (nmol/mol);

hv is the hourly alert threshold value (nmol/mol);

$s_{r,f}$ is the standard uncertainty for reproducibility under field conditions (%).

The standard uncertainty due the **long term drift at zero**, $u_{d,l,z}$, is calculated according to:

$$u_{d,l,z} = D_{l,z} / \sqrt{3} \quad (G.35)$$

where

$u_{d,l,z}$ is the uncertainty due to long term drift at zero (nmol/mol);

$D_{l,z}$ is the long term drift at zero (nmol/mol).

The standard uncertainty due to the **long term drift at level of the hourly alert threshold value**, u_{d,l,l_v} , is calculated according to:

$$u_{d,l,l_v} = (D_{l,l_v} / 100) \times hlv / \sqrt{3} \quad (\text{G.36})$$

where

u_{d,l,l_v} is the uncertainty due to long term drift at the hourly alert threshold value (nmol/mol);

D_{l,l_v} is the long term drift at level of the hourly alert threshold value (%);

hlv is the hourly alert threshold value (nmol/mol).

The standard uncertainty due to the **calibration gas**, u_{cg} , is calculated according to Equation (G.31).

The standard uncertainty due to the **difference sample/calibration port**, u_{Dsc} , is calculated according to Equation (G.32).

G.7 Example Result of type approval for O₃ with the results of the laboratory and field tests

G.7.1 Values of test parameters to be used in the uncertainty calculations

Next to the values given in G.4.1 the following values are used.

Table G.18 — Test concentration of the measurand

Average concentration over period of long term drift	20	nmol/mol
--	----	----------

Table G.19 — Default value calibration frequency

Calibration frequency	1 × / 3 months
-----------------------	----------------

G.7.2 Uncertainty due to interferences, except water vapour (see EN ISO 14956:2002, 8.5.5)

See G.4.2.

G.7.3 Calculation of expanded uncertainty from the results of laboratory and field tests for O₃

Table G.20 — Uncertainty from the results of laboratory and field tests

No.	Standard uncertainty due to	Result or default value		Equation G.	uncert. u(p) nmol/mol	uncert. u(p) ² (nmol/mol) ²
1	Repeatability standard deviation at zero	0,45	nmol/mol	6, 7	0,05	0,00
2	Repeatability standard deviation at the hourly alert threshold value	2,4	nmol/mol	8, 9, 10	0,18 ^b	0,03
3	Lack of fit at the hourly alert threshold value	2,5	% of the tested value	11	1,73	3,00
4	Sensitivity coefficient of sample gas pressure	0,20	nmol/mol/kPa	12, 13	2,19	4,79
5	Sensitivity coefficient of sample gas temperature	0,35	nmol/mol/K	14, 15	3,83	14,7
6	Sensitivity coefficient of surrounding temperature	0,20	nmol/mol/K	16, 17	2,19	4,79
7	Sensitivity coefficient of electrical voltage	0,25	nmol/mol/V	18, 19	3,65	13,3
8	Interferents at level of the hourly alert threshold value					
	8a H ₂ O with concentration 21 mmol/mol ^a	-3,2	nmol/mol	20, 21, 22, 23	-2,23	4,99
	8b-8c summed uncertainty with positive or negative impact (whichever is the greatest)			24, 25, 26, 27, 28, 29	0,60	0,36
9	Averaging error	5,0	% of the tested value	30	3,46	12,0
18	Difference sampl/calibration port	0,70	%	32	0,48	0,24
10	Reproducibility standard deviation under field conditions	1,7	% of average of 3-months period	34	2,04 ^b	4,16
11	Long term drift at zero	0,99	nmol/mol	35	0,57	0,33
12	Long term drift at limit value	3,2	% of limit value	36	2,22	4,92
21	Uncertainty in calibration gas	3,0	% of limit value	31	1,80	3,24
^a	calculated by extrapolation from the test results at 19 mmol/mol (see Equations G.20, G.21, G.22, G.23)					
^b	Take for the calculation of the combined standard uncertainty the uncertainty due to the repeatability at the hourly alert threshold value or the uncertainty due to the reproducibility under field conditions, whichever is the greatest.					

Table G.21 — Result of the calculation

combined standard uncertainty	8,41	nmol/mol
expanded uncertainty	16,8	nmol/mol
expanded uncertainty actual	14,0	%
expanded uncertainty required	15	%

CONCLUSION: $U_{c, \text{rel}} \leq U_{\text{req, rel}}$

Requirement 4 is fulfilled.

Annex H (normative)

Calculation of uncertainty in field operation at the hourly alert threshold value

H.1 Uncertainty in hourly measurements at the level of the hourly alert threshold value

H.1.1 Combined standard uncertainty

The **combined standard uncertainty**, u_c , in an hourly measurement at the hourly alert threshold value during field operation of a measurement system shall be calculated using the following equation:

$$u_c = \sqrt{u_{r,z}^2 + (u_{r,lv} \text{ or } u_{r,f})^2 + u_l^2 + u_{gp}^2 + u_{gt}^2 + u_{st}^2 + u_v^2 + u_{H_2O,act}^2 + (u_{int,act,pos} \text{ or } u_{int,act,neg})^2 + u_{av}^2 + u_{Dsc}^2 + u_{d,l,z}^2 + u_{d,l,lv}^2 + u_{cg}^2} \quad (H.1)$$

where

- $u_{r,z}$ is the standard uncertainty for repeatability at zero (nmol/mol);
- $u_{r,lv}$ is the standard uncertainty for repeatability at the hourly alert threshold value (nmol/mol). The highest value of $u_{r,lv}$ and $u_{r,f}$ shall be taken in the uncertainty calculation.
- u_l is the standard uncertainty for lack of fit at the hourly alert threshold value (nmol/mol);
- u_{gp} is the standard uncertainty for sample gas pressure variation (nmol/mol);
- u_{gt} is the standard uncertainty for sample gas temperature variation (nmol/mol);
- u_{st} is the standard uncertainty for surrounding temperature variation (nmol/mol);
- u_v is the standard uncertainty for electrical voltage variation (nmol/mol);
- $u_{H_2O,act}$ is the standard uncertainty for the presence of water vapour (nmol/mol);
- $u_{int,act,pos}$ is the standard uncertainty for interferents (except water vapour) with positive impact (nmol/mol). The highest value of $u_{int,act,pos}$ and $u_{int,act,neg}$ shall be taken in the uncertainty calculation;
- $u_{int,act,neg}$ is the standard uncertainty for interferents (except water vapour) with negative impact (nmol/mol). The highest value of $u_{int,act,pos}$ and $u_{int,act,neg}$ shall be taken in the uncertainty calculation;
- u_{av} is the standard uncertainty for averaging (nmol/mol);
- u_{Dsc} is the standard uncertainty for difference sample/calibration port (nmol/mol);
- $u_{r,f}$ is the standard uncertainty for reproducibility under field conditions (nmol/mol). The highest value of $u_{r,lv}$ and $u_{r,f}$ shall be taken in the uncertainty calculation;
- $u_{d,l,z}$ is the standard uncertainty for long-term drift at zero (nmol/mol);
- $u_{d,l,lv}$ is the standard uncertainty for long-term drift at level of the hourly alert threshold value (nmol/mol);
- u_{cg} is the standard uncertainty for calibration gas (nmol/mol).

The absolute **expanded uncertainty**, U_c , shall be calculated according to:

$$U_c = k \times u_c \quad (\text{H.2})$$

with $k = 2$

where

- U_c is the expanded uncertainty (nmol/mol);
- k is the coverage factor of approximately 95 %;
- u_c is the combined standard uncertainty (nmol/mol).

The relative **expanded uncertainty**, $U_{c,rel}$, shall be calculated according to:

$$U_{c,rel} = (U_c / hlv) \times 100 \quad (\text{H.3})$$

where

- $U_{c,rel}$ is the relative expanded uncertainty (%);
- U_c is the combined expanded uncertainty (nmol/mol);
- hlv is the hourly alert threshold value (nmol/mol);
- U_c is the expanded uncertainty (nmol/mol);
- hlv is the hourly alert threshold value (nmol/mol).

H.1.2 Standard uncertainties

The standard uncertainties shall be calculated with the equations given in Annex G, using the relevant values of the performance characteristics, the values of the site-specific conditions related to physical and chemical influences, the value of the site-specific conditions related to operational parameters and the actual values of test gas concentrations during the type approval test.

The standard uncertainty for **repeatability at zero**, $u_{r,z}$, is calculated according to Equations (G.6, G.7).

The standard uncertainty for **repeatability at the hourly alert threshold value**, $u_{r,lv}$, is calculated according to Equations (G.8, G.9, G.10)

The standard uncertainty due to **lack of fit at the hourly alert threshold value**, u_l , is calculated according to Equation (G.11).

The standard uncertainty due to variation of **sample gas pressure at the hourly alert threshold value**, u_{sp} , is calculated according to:

$$u_{gp} = ((hlv/c_t) \times b_{gp}) (\Delta_{gp} / \sqrt{3}) \quad (\text{H.4})$$

$$\Delta_{gp} = P_1 - P_2 \quad (\text{H.5})$$

where

- u_{gp} is the standard uncertainty due to the influence of pressure (nmol/mol);
- hlv is the hourly alert threshold value (nmol/mol);

- c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);
- b_{gp} is the sensitivity coefficient of sample gas pressure variation (nmol/mol/kPa);
- Δ_{gp} is the pressure range of the variation of the sample gas pressure (kPa);
- P_1 is the upper level of the site-specific range of the variation of the sample gas pressure (kPa);
- P_2 is the lower level of the site-specific range of the variation of the sample gas pressure (kPa).

The standard uncertainty due to variation of **sample gas temperature at the hourly alert threshold value**, u_{gt} , is calculated according to:

$$u_{gt} = ((h/v / c_t) \times b_{gt})(\Delta_{gt} / \sqrt{3}) \quad (H.6)$$

$$\Delta_{gt} = T_{gt,1} - T_{gt,2} \quad (H.7)$$

where

- u_{gt} is the standard uncertainty due to the variation of sample gas temperature (nmol/mol);
- h/v is the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);
- b_{gt} is the sensitivity coefficient of sample gas temperature variation (nmol/mol/K);
- Δ_{gt} is the sample gas temperature range of the variation of the sample gas temperature (K);
- $T_{gt,1}$ is the upper level of the site-specific range of the variation of the sample gas temperature (°C);
- $T_{gt,2}$ is the lower level of the site-specific range of the variation of the sample gas temperature (°C).

The standard uncertainty due to variation of **surrounding temperature at the hourly alert threshold value**, u_{st} , is calculated according to:

$$u_{st} = ((h/v / c_t) \times b_{st})(\Delta_{st} / \sqrt{3}) \quad (H.8)$$

$$\Delta_{st} = T_{st,1} - T_{st,2} \quad (H.9)$$

where

- u_{st} is the standard uncertainty due to the variation of the surrounding temperature (nmol/mol);
- h/v is the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);
- b_{st} is the sensitivity coefficient of the surrounding temperature variation (nmol/mol/K);
- Δ_{st} is the surrounding temperature range of the variation of the surrounding temperature (K);
- $T_{st,1}$ is the upper level of the site-specific range of the variation of the surrounding temperature (°C) (see H.2.3);
- $T_{st,2}$ is the lower level of the site-specific range of the variation of the surrounding temperature (°C) (see H.2.3).

The standard uncertainty due to variation of **electrical voltage at the hourly alert threshold value**, u_v , is calculated according to:

$$u_v = ((h/v / c_t) \times b_v) (\Delta_v / \sqrt{3}) \quad (\text{H.10})$$

$$\Delta_v = V_1 - V_2 \quad (\text{H.11})$$

where

- u_v is the standard uncertainty due to the variation of electrical voltage (nmol/mol);
- h/v is the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration (70 % to 80 % of the certification range) (nmol/mol);
- b_v is the sensitivity coefficient of electrical voltage variation (nmol/mol/V);
- Δ_v is the electrical voltage range of the variation of the electrical voltage (V);
- V_1 is the upper level of the site-specific range of the variation of the electrical voltage (V);
- V_2 is the lower level of the site-specific range of the variation of the electrical voltage (V).

The calculation of the uncertainty due to interferences has to be based on the actual concentration of the chemical interferences during field operation. Therefore the following equations have to be used for calculating the uncertainty due to water vapour and other chemical interfering compounds

The standard uncertainty due to the **presence of water vapour at the hourly alert threshold value**, $u_{\text{H}_2\text{O,act}}$, is calculated according to:

$$X_{\text{H}_2\text{O,z,act}} = (c_{\text{H}_2\text{O,act,max}} / c_{\text{H}_2\text{O,max}}) \times X_{\text{H}_2\text{O,z,max}} \quad (\text{H.12})$$

$$X_{\text{H}_2\text{O,c}_t,\text{act}} = (c_{\text{H}_2\text{O,act,max}} / c_{\text{H}_2\text{O,max}}) \times X_{\text{H}_2\text{O,c}_t,\text{max}} \quad (\text{H.13})$$

$$X_{\text{H}_2\text{O,act}} = ((X_{\text{H}_2\text{O,c}_t,\text{act}} - X_{\text{H}_2\text{O,z,act}}) / c_t) \times h/v + X_{\text{H}_2\text{O,z,act}} \quad (\text{H.14})$$

$$u_{\text{H}_2\text{O,act}} = |(X_{\text{H}_2\text{O,act}} / c_{\text{H}_2\text{O,act,max}})| \times \sqrt{((c_{\text{H}_2\text{O,act,max}})^2 + c_{\text{H}_2\text{O,act,max}} \times c_{\text{H}_2\text{O,act,min}} + c_{\text{H}_2\text{O,act,min}}^2) / 3} \quad (\text{H.15})$$

where

- $X_{\text{H}_2\text{O,z,act}}$ is the influence quantity of the actual H_2O concentration at zero concentration of the measurand (nmol/mol);
- $X_{\text{H}_2\text{O,c}_t,\text{act}}$ is the influence quantity of the actual H_2O concentration at the test concentration c_t of the measurand (nmol/mol);
- $c_{\text{H}_2\text{O,max}}$ is the maximum concentration of water vapour used in the calculations for type approval (= 21 mmol/mol);
- $X_{\text{H}_2\text{O,z,max}}$ is the influence quantity of an H_2O concentration of 121 mmol/mol at zero concentration of the measurand (nmol/mol) (calculated with Equation G.20);
- $X_{\text{H}_2\text{O,c}_t,\text{max}}$ is the influence quantity of an H_2O concentration of 21 mmol/mol at the test concentration c_t of the measurand (nmol/mol) (calculated with Equation G.21);
- $X_{\text{H}_2\text{O,act}}$ is the influence quantity of the actual H_2O concentration at the hourly alert threshold value (nmol/mol);
- c_t is the test gas concentration of the measurand (nmol/mol);

h/v	is the hourly alert threshold value (nmol/mol);
$u_{H_2O,act}$	is the standard uncertainty due to interference by the presence of water vapour at the actual concentration of water vapour (nmol/mol);
$c_{H_2O,act,max}$	is the actual maximum concentration of water vapour (mmol/mol);
$c_{H_2O,act,min}$	is the actual minimum concentration of water vapour (mmol/mol).

The standard uncertainty due to each **interfering compound at the hourly alert threshold value** (other than water vapour), $u_{int,act}$, is calculated according to:

$$X_{int,z,act} = (c_{int,act,max} / c_{int,max}) \times X_{int,z,max} \quad (H.16)$$

$$X_{int,c_t,act} = (c_{int,act,max} / c_{int,max}) \times X_{int,c_t,max} \quad (H.17)$$

$$X_{int,act} = ((X_{int,c_t,act} - X_{int,z,act}) / c_t) \times h/v + X_{int,z,act} \quad (H.18)$$

$$u_{int,act} = \left| (X_{int,act} / c_{int,act,max}) \right| \times \sqrt{((c_{int,act,max}^2 + c_{int,act,max} \times c_{int,act,min} + c_{int,act,min}^2) / 3)} \quad (H.19)$$

where

$X_{int,act,z}$	is the influence quantity of the actual concentration (nmol/mol resp. μ mol/mol) of the relevant interfering compound at zero concentration of the measurand (nmol/mol);
$X_{int,act,ct}$	is the influence quantity of the actual concentration (nmol/mol resp. μ mol/mol) of the relevant interfering compound at the test concentration c_t of the measurand (nmol/mol);
$c_{int,max}$	is the maximum concentration of the relevant interfering compound used in the calculations for type approval (nmol/mol);
$X_{int,max,z}$	is the influence quantity of the maximum concentration (nmol/mol resp. μ mol/mol) of the relevant interfering compound at zero concentration of the measurand (nmol/mol);
$X_{int,max,ct}$	is the influence quantity of the maximum concentration (nmol/mol resp. μ mol/mol) of the relevant interfering compound at the test concentration c_t of the measurand (nmol/mol);
c_t	is the test concentration of the measurand at the level of the hourly alert threshold (nmol/mol);
h/v	is the hourly alert threshold value (nmol/mol);
$u_{int,act}$	is the standard uncertainty at the hourly alert threshold value due to interference by the presence of a chemical compound at the actual concentration of the compound (nmol/mol);
$c_{int,act,max}$	is the actual maximum concentration of the relevant interfering compound (nmol/mol resp. μ mol/mol);
$c_{int,act,min}$	is the actual minimum concentration of the relevant interfering compound (nmol/mol resp. μ mol/mol);

According to EN ISO 14956 the summed uncertainties due to the interferents with positive impact and the summed uncertainties of the interferents with negative impact shall be calculated:

$$S(u_{int,act,pos}) = u_{int,act,1,pos} + u_{int,act,2,pos} + \dots + u_{int,act,n,pos} \quad (H.20)$$

$$S(u_{\text{int},1,\text{neg}}) = u_{\text{int},1,\text{neg}} + u_{\text{int},2,\text{neg}} + \dots + u_{\text{int},n,\text{neg}} \quad (\text{H.21})$$

Take the highest sum as the representative value for all interferents:

$$u_{\text{int},\text{act},\text{pos}} = \sqrt{S(u_{\text{int},\text{act},\text{pos}})^2} \quad (\text{H.22})$$

$$u_{\text{int},\text{act},\text{neg}} = \sqrt{S(u_{\text{int},\text{act},\text{neg}})^2} \quad (\text{H.23})$$

where

$u_{\text{int},\text{act},\text{pos}}$ is the sum of uncertainties due to interferents with positive impact (nmol/mol);

$u_{\text{int},\text{act},1,\text{pos}}$ is the uncertainty due to the first interferent with positive impact (nmol/mol);

$u_{\text{int},\text{act},n,\text{pos}}$ is the uncertainty due to the n th interferent with positive impact (nmol/mol);

$u_{\text{int},\text{act},\text{neg}}$ is the sum of uncertainties due to interferents with negative impact (nmol/mol);

$u_{\text{int},\text{act},1,\text{neg}}$ is the uncertainty due to the first interferent with negative impact (nmol/mol);

$u_{\text{int},\text{act},n,\text{neg}}$ is the uncertainty due to the n th interferent with negative impact (nmol/mol);

The standard uncertainty due to the **averaging error**, u_{av} , is calculated according to Equation (G.30).

The standard uncertainty due to the **reproducibility under field conditions**, $u_{\text{r,f}}$, is calculated according to Equation (G.34).

The standard uncertainty due the **long term drift at zero**, $u_{\text{d,l,z}}$, is calculated according to Equation (G.35):

The standard uncertainty due to the **long term drift at level of the hourly alert threshold value**, $u_{\text{d,l,lv}}$, is calculated according to Equation (G.32).

The standard uncertainty due to the **calibration gas**, u_{cg} , is calculated according to Equation (G.31), with the actual uncertainty of the calibration gas.

The standard uncertainty due to the **difference sample/calibration port**, u_{Dsc} , is calculated according to Equation (G.32).

H.2 Example

H.2.1 General

To calculate the uncertainty at the hourly alert threshold value during field operation of a measurement system the following data shall be used:

- values of the following performance characteristics obtained during the type approval test:
 - Repeatability standard deviation at zero;
 - Repeatability standard deviation at test gas concentration c_{t} ;
 - Lack of fit;

- Sensitivity coefficient of sample gas pressure;
- Sensitivity coefficient of sample gas temperature;
- Sensitivity coefficient of surrounding temperature;
- Sensitivity coefficient of electrical voltage;
- Interferences at zero and test gas concentration c_i ;
 - H_2O with concentration 19 mmol/mol;
 - toluene with concentration 0,5 $\mu\text{mol/mol}$;
 - xylene with concentration 0,5 $\mu\text{mol/mol}$;
- Averaging error;
- Difference sample/calibration port (if relevant);
- Reproducibility standard deviation under field conditions;
- Long term drift at zero;
- Long term drift at span level;
- values of site specific conditions related to physical influences:
 - Sample pressure variation;
 - Sample gas temperature variation;
 - Surrounding air temperature variation;
 - Voltage variation;
- values of site specific conditions related to chemical influences/interferents:
 - H_2O concentration range;
 - toluene concentration range;
 - xylene concentration range;
- values of site specific conditions related to operational parameters:
 - Uncertainty in calibration gas;
 - Calibration frequency;
- actual values of test gas concentrations during the type approval test.

H.2.2 Values of the relevant performance characteristics obtained during the type approval test

Table H.1 gives the results of the type approval, both from the laboratory and the field tests, to be incorporated in the uncertainty calculation, together with the relevant performance criteria.

Table H.1 — Relevant performance characteristics and criteria

No.	Performance characteristic	Symbol	Clause	Performance criterion for O ₃	Results lab. and field test
					^a
1	Repeatability standard deviation at zero	$s_{r,z}$	8.4.5	$\leq 1,0$ nmol/mol	0,45
2	Repeatability standard deviation at concentration c_t (at a level of the hourly alert threshold)	$s_{r,ct}$	8.4.5	$\leq 3,0$ nmol/mol	2,4
3	Lack of fit (at the hourly alert threshold)	X_l	8.4.6	$\leq 5,0$ % of the measured value	2,5
4	Sensitivity coefficient of sample gas pressure	b_{gp}	8.4.7	$\leq 2,0$ nmol/mol/kPa	0,20
5	Sensitivity coefficient of sample gas temperature	b_{gt}	8.4.8	$\leq 1,0$ nmol/mol/K	0,35
6	Sensitivity coefficient of surrounding temperature	b_{st}	8.4.9	$\leq 1,0$ nmol/mol/K	0,20
7	Sensitivity coefficient of electrical voltage	b_v	8.4.10	$\leq 0,3$ nmol/mol/V 0	0,25
8	Interferents at zero and concentration c_t (at a level of the hourly alert threshold) ^b		8.4.11		
8a	H ₂ O with concentration 19 mmol/mol ^c	$X_{H_2O,z}$		≤ 10 nmol/mol	-2,3
		$X_{H_2O,ct}$		≤ 10 nmol/mol	-2,9
8b	toluene with concentration 0,5 μ mol/mol	$X_{benz,z}$		$\leq 5,0$ nmol/mol	0,49
		$X_{benz,ct}$		$\leq 5,0$ nmol/mol	0,57
8c	xylene with concentration 0,5 μ mol/mol	$X_{benz,z}$		$\leq 5,0$ nmol/mol	0,39
		$X_{benz,ct}$		$\leq 5,0$ nmol/mol	0,47
9	Averaging error	X_{av}	8.4.12	$\leq 7,0$ % of the measured value	5,0
18	Difference sample/calibration port ^d	D_{sc}	8.4.13	$\leq 1,0$ %	0,70
10	Reproducibility standard deviation	$s_{r,f}$	8.5.5	$\leq 5,0$ % of the average of a	1,7

No.	Performance characteristic	Symbol	Clause	Performance criterion for O ₃	Results lab. and field test
	under field conditions			three month period	
11	Long term drift at zero	$D_{l,z}$	8.5.4	$\leq 5,0$ nmol/mol	0,99
12	Long term drift at span level	$D_{l,s}$	8.5.4	$\leq 5,0$ % of maximum of certification range	3,2
15	Response time (rise)	t_r	8.4.3	≤ 180 s	55
16	Response time (fall)	t_f	8.4.3	≤ 180 s	50
^a to demonstrate compliance the absolute value of the performance characteristic shall be taken ^b performance criterion is set both at zero and span level ^c an H ₂ O-concentration of 19 mmol/mol equals 80 % RH at 20 °C and 101,3 kPa ^d if relevant					
NOTE nmol/mol = ppb μ mol/mol = ppm					

H.2.3 Values of site specific conditions related to physical influences

Table H.2 — Site specific conditions related to physical influences

	site specific			type approval		
	lower value	upper value	range	lower value	upper value	test range
Sample pressure variation	95 kPa	110 kPa	15 kPa ^a	80 kPa	110 kPa	30 kPa
Sample gas temperature variation	0 °C	30 °C	30 K	0 °C	30 °C	30 K
Surrounding air temperature variation	10 °C	30 °C	20 K	0 °C	30 °C	30 K
Voltage variation	220 V	250 V	30 V	210 V ^a	250 V ^a	40 V ^a

H.2.4 Values of site-specific conditions related to chemical influences/interferents

Table H.3 — Site-specific conditions related to chemical influences/interferents

Interferents		site specific (hourly average values)			type approval	
		to be used in uncertainty calculation during operation			used in uncertainty calculation for type approval	
		lower value, $c_{\text{int,act,min}}$	upper value, $c_{\text{int,act,max}}$		lower value, $c_{\text{int,min}}$	upper value, $c_{\text{int,max}}$
H ₂ O	mmol/mol	3	21		6	21
toluene	μmol/mol	0	0,5		0	0,5
xylene	μmol/mol	0	0,5		0	0,5

H.2.5 Values of site specific conditions related to operational parameters

Table H.4 — Site specific conditions related to operational parameters

	site specific	Type approval
Uncertainty in calibration gas, expressed as 2 times the standard deviation	± 5 %	± 3 %
Calibration frequency	1x / 3 months	1x / 3 months

H.2.6 Actual values of test gas concentrations during the type approval test

Table H.5 — Test concentrations of the measurand

c_i at a level of the hourly alert threshold value	120	nmol/mol
c_i at span level (70 % to 80 % of the certification range)	190	nmol/mol
average concentration over period of long term drift	20	nmol/mol

H.2.7 Result of uncertainty calculation for O₃ with the results of the type approval tests and the site specific conditions

Table H.6 - Result of uncertainty calculation for O₃

N o.	Performance characteristic	Equation	Partial uncertainty	Value of partial uncertainty nmol/mol
1	Repeatability standard deviation at zero	G.6, 7	$u_{r,z}$	$(0,45 / \sqrt{(69)}) = 0,03$
2	Repeatability at hourly alert threshold	G.8, 9, 10	$u_{r,lv}$	$((120 / 125) \cdot 2,4) / \sqrt{(69)} = 0,28$
3	Lack of fit	G.11	u_l	$((2,50 / 100) \cdot 120) / \sqrt{3} = 1,73$
4	Sensitivity coefficient of sample gas pressure	H.4, 5	u_{gp}	$((120 / 190) \cdot 0,20 \cdot 15) / \sqrt{3} = 1,04$
5	Sensitivity coefficient of sample gas temperature	H.6, 7	u_{gt}	$((120 / 190) \cdot 0,35 \cdot 30) / \sqrt{3} = 1,91$

N o.	Performance characteristic	Equation	Partial uncertainty	Value of partial uncertainty nmol/mol
6	Sensitivity coefficient of surrounding temperature	H.8, 9	u_{st}	$((120 / 190) \cdot 0,20 \cdot 20) / \sqrt{3} = 0,73$
7	Sensitivity coefficient of electrical voltage	H.10, 11	u_v	$((120 / 190) \cdot 0,25 \cdot 30) / \sqrt{3} = 1,87$
8	Interferences			
8a	H ₂ O with actual concentration 21 mmol/mol	H.12, 13, 14, 15	$u_{H_2O,act}$	$\left \frac{((18/18) \cdot ((18/16) \cdot (-2,9)) - (18/18) \cdot ((18/16) \cdot (-2,3)) \cdot 120 + (18/18) \cdot ((18/16) \cdot (-2,9))}{125} \right \cdot 18$ $\sqrt{\frac{18^2 + 18 \cdot 3 + 3^2}{3}} = 2,04$
8b	toluene with concentration 0,5 μmol/mol	H.16, 17, 18, 19	$u_{tol,act}$	$\frac{((0,5/0,5) \cdot 0,57 - ((0,5/0,5) \cdot 0,49)) \cdot 120 + ((0,5/0,5) \cdot 0,49))}{125} \cdot 0,5$ $\sqrt{\frac{0,5^2 + 0,5 \cdot 0 + 0^2}{3}} = 0,33$
8b	xylene with concentration 0,5 μmol/mol	H. 16, 17, 18, 19	$u_{xyl,act}$	$\frac{((0,5/0,5) \cdot 0,47 - ((0,5/0,5) \cdot 0,39)) \cdot 120 + ((0,5/0,5) \cdot 0,39))}{125} \cdot 0,5$ $\sqrt{\frac{0,5^2 + 0,5 \cdot 0 + 0^2}{3}} = 0,33$
9	Averaging error	G.30	u_{av}	$((5,0 / 100) \cdot 120) / \sqrt{3} = 3,46$
18	Difference sample/calibration port	G.32	u_{Dsc}	$(0,70 / 100) \cdot 120 / \sqrt{3} = 0,48$
10	Reproducibility standard deviation under field conditions	G.34	$u_{r,f}$	$(1,7 / 100) \cdot 120 = 2,04$
11	Long term drift at zero	G.35	$u_{d,l,z}$	$0,99 / \sqrt{3} = 0,57$
12	Long term drift at span level	G.36	$u_{d,l,lv}$	$((3,2 / 100) \cdot 120) / \sqrt{3} = 2,22$
20	Uncertainty in calibration gas	G.31	u_{cg}	$((5,0 / 100) \cdot 120) / 2 = 3,00$

Compare the values of $u_{r,lv}$ and $u_{r,f}$. Take the highest one in the calculation of the combined uncertainty:

$$u_{r,lv} = 0,28$$

$$u_{r,f} = 2,04$$

According to EN ISO 14956 the summed uncertainties due to the interferences with positive impact and the summed uncertainties of the interferences with negative impact shall be calculated. Take the highest sum as the representative value for all interferences (see Equations (G.18, G.19) and (G.20, G.21, G.22, G.23)).

$$u_{\text{int,pos}} = 0,60$$

$$u_{\text{int,neg}} = 0,00$$

Calculate the combined uncertainty (at the hourly alert threshold) with Equation (G.3):

$$u_c = \sqrt{(0,03^2 + 1,73^2 + 1,04^2 + 1,91^2 + 0,73^2 + 1,87^2 + 1,98^2 + (0,33 + 0,33)^2 + 3,46^2 + 0,48^2 + 2,04 + 0,57^2 + 2,22^2 + 3,00^2)} = 6,84 \text{ nmol/mol}$$

The absolute **expanded uncertainty**, U_c , shall be calculated according to Equation (G.4):

$$U_c = 2 \cdot 6,84 = 13,68 \text{ nmol/mol}$$

The relative **combined expanded uncertainty**, $U_{c,\text{rel}}$, shall be calculated according to Equation (G.5):

$$U_{c,\text{rel}} = (13,68 / 120) \cdot 100 = 11,4 \%$$

The requirement is: $U_{c,\text{rel}} \leq U_{\text{req,rel}}$

$$U_{c,\text{rel}} = 11,4 \%$$

$$U_{\text{req,rel}} = 15 \%$$

The requirement is fulfilled.

Bibliography

- [1] VIM:1993, *International vocabulary of basic and general terms in metrology*, 2nd edition, ISO
- [2] CR 14377:2002, *Air quality – Approach to uncertainty estimation for ambient air reference measurement methods*
- [3] C.P. Meyer, C.M. Elsworth and I.E. Galbally, *Water vapor interference in the measurement of ozone in ambient air by ultraviolet absorption*, *Rev. Sci. Instrum.*, **62** (1), 223 (1991).
- [4] ISO 10313, *Ambiant air - Determination of the mass concentration of ozone - Chemiluminescence method*
- [5] D. Daumont, J.Brion, J. Charbonnier and J. Malicet, *Ozone UV Spectroscopy I: Absorption Crosssections at Room Temperature*, *J. Atmos. Chem.*, **15**, 145 (1992).
- [6] *Technical Assistance Document for the Calibration of Ambient Ozone Monitors*, Report EPA-600/4-79-057, September 1979, United States Environmental Protection Agency, Research Triangle Park, N.C. 27711, USA. 8] *Transfer Standards for Calibration of Air Monitoring Analysers for Ozone*, Report EPA-600/4-79-056, September 1979, United States Environmental Protection Agency, Research Triangle Park, N.C. 27711, USA.
- [7] *Standard Practices for Calibration of Ozone Monitors Using Transfer Standards*, American Society for Testing and Materials, ASTM Standard: D 5011-92 (1992).
- [8] *Standard Practice for Calibration of Ozone Monitors and Certification of Ozone Transfer Standards Using Ultraviolet Photometry*, American Society for Testing and Materials, ASTM Standard: D 5110-90 (1991).
- [9] Verein Deutscher Ingenieure, *Messen der Ozonkonzentration Direktes UV-photometrisches Verfahren (Basisverfahren)*, VDI 2468, Blatt 6 (1979).
- [10] AFNOR (Association Française de Normalisation), *Dosage de l'ozone dans l'air - Methode par absorption UV*, X43-024 (1986).
- [11] Verein Deutscher Ingenieure, *Messen der Ozon- und Peroxid- Konzentrationen: Manuelles photometrisches Verfahren Kaliumjodid-Methode (Basisverfahren)*, VDI 2468, Blatt 1 (1978).
- [12] *Evaluation of Ozone Calibration Procedures*, Report EPA-600/S4-80-050, February 1981, United States Environmental Protection Agency, Research Triangle Park, N.C. 27711, USA.
- [13] D. Grosjean and J. Harrison, *Response of Chemiluminescence NOX Analysers and Ultraviolet Ozone Analysers to Organic Air Pollutants*, *Envir. Sci. Technol.*, **19**, 862 (1985).
- [14] T. Kleindienst, E.E. Hudgens, D.G. Smith, G.G. McElroy and J.J. Bufalini, *Comparison of Chemiluminescence and Ultraviolet Ozone Monitor Responses in the Presence of Humidity and Photochemical Pollutants*, *J. Air Wast Manage. Assoc.*, **43**, 213 (1993).
- [15] ISO 4226, *Air quality - General aspects - Units of measurement*
- [16] ISO 6879, *Air quality - Performance characteristics and related concepts for air quality measuring methods*
- [17] Harmonisation of directive 92/72/EEC on air pollution by ozone inter-comparisons of calibration procedures for ozone measurements 3 to 7 October 1994 – 20 to 24 March 1995, Ispra, Italy, IRC, EUR 17662 EN, 1997
- [18] DIRECTIVE 2002/3/EC of the European Parliament and of the council of 12 February 2002 relating to ozone in ambient air

- [19] EN ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories* (ISO/IEC 17025:1999)

BSI — British Standards Institution

BSI is the independent national body responsible for preparing British Standards. It presents the UK view on standards in Europe and at the international level. It is incorporated by Royal Charter.

Revisions

British Standards are updated by amendment or revision. Users of British Standards should make sure that they possess the latest amendments or editions.

It is the constant aim of BSI to improve the quality of our products and services. We would be grateful if anyone finding an inaccuracy or ambiguity while using this British Standard would inform the Secretary of the technical committee responsible, the identity of which can be found on the inside front cover.
Tel: +44 (0)20 8996 9000. Fax: +44 (0)20 8996 7400.

BSI offers members an individual updating service called PLUS which ensures that subscribers automatically receive the latest editions of standards.

Buying standards

Orders for all BSI, international and foreign standards publications should be addressed to Customer Services. Tel: +44 (0)20 8996 9001.
Fax: +44 (0)20 8996 7001. Email: orders@bsi-global.com. Standards are also available from the BSI website at <http://www.bsi-global.com>.

In response to orders for international standards, it is BSI policy to supply the BSI implementation of those that have been published as British Standards, unless otherwise requested.

Information on standards

BSI provides a wide range of information on national, European and international standards through its Library and its Technical Help to Exporters Service. Various BSI electronic information services are also available which give details on all its products and services. Contact the Information Centre.
Tel: +44 (0)20 8996 7111. Fax: +44 (0)20 8996 7048. Email: info@bsi-global.com.

Subscribing members of BSI are kept up to date with standards developments and receive substantial discounts on the purchase price of standards. For details of these and other benefits contact Membership Administration.
Tel: +44 (0)20 8996 7002. Fax: +44 (0)20 8996 7001.
Email: membership@bsi-global.com.

Information regarding online access to British Standards via British Standards Online can be found at <http://www.bsi-global.com/bsonline>.

Further information about BSI is available on the BSI website at <http://www.bsi-global.com>.

Copyright

Copyright subsists in all BSI publications. BSI also holds the copyright, in the UK, of the publications of the international standardization bodies. Except as permitted under the Copyright, Designs and Patents Act 1988 no extract may be reproduced, stored in a retrieval system or transmitted in any form or by any means – electronic, photocopying, recording or otherwise – without prior written permission from BSI.

This does not preclude the free use, in the course of implementing the standard, of necessary details such as symbols, and size, type or grade designations. If these details are to be used for any other purpose than implementation then the prior written permission of BSI must be obtained.

Details and advice can be obtained from the Copyright & Licensing Manager.
Tel: +44 (0)20 8996 7070. Fax: +44 (0)20 8996 7553.
Email: copyright@bsi-global.com.